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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Introduction Chapter (1): Not started. Will write last/almost last
- IFO Chapter (2): Apparently this is a good chapter...
- PSL Chapter (3): Passed first drafting. Very Minor corrections to be made.
- BHD Chapter (5): Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
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- SRM beam parameter Chapter (4): Still to apply Ken and Bryan's comments
- MZ chapter Chapter (??): Not written, vaguely planned
- Photodiode Chapter (8): Need to finish adding references.
- Conclusion Chapter (??): Not started. Will write last/almost last

Chapter 1

Introduction

Chapter 2

The Fundamentals of Gravitational Wave Interferometers

2.1 Introduction

On the 14th of September 2015, the detection of a gravitational wave emanating from two coalescing black holes signalled the beginning of a new era for astronomy [1]. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. Gravitational wave astronomy also allows for direct observations of objects that do not emit electromagnetic radiation, e.g. black holes. Therefore, it is important that large numbers of gravitational wave signals are measured to develop our understanding of the universe.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferometer, with a particular emphasis on LIGO [2], can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the sensitivity required to detect gravitational waves; therefore, quantum noise is the focus of this chapter.

2.2 The Astrophysical Origins of Gravitational Waves

Gravitational waves produced by inspiraling pairs of massive objects, such as binary black holes (BBH) (e.g. [1, 3–5]), binary neutron stars (BNS) [6, 7], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [8], have been detected. Every confirmed gravitational wave signal to date is summarised in [9]. The waveform of the gravitational radiation emitted from these systems is known as a chirp because the frequency of the gravitational wave signal increases as the two objects get closer to merging [10]. The cut-off frequency is determined by the radius of the objects that will merge; for the signals which have been measured, the cut-off frequencies were between 100 Hz and 10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein’s general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo interferometers [11]. Gravitational waves are

emitted from mass quadrupole systems (see e.g. [12] or [13]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses moving at relativistic velocities.

Other astrophysical phenomena will emit gravitational waves. Asymmetries in pulsars [14,15] and magnetars [16] will cause them to emit a gravitational wave with a constant frequency. When the core of a star collapses as it goes supernova, a burst of gravitational radiation may be emitted [17,18]. Burst signals may also be generated immediately after a BNS merges [19]. Weaker sources that future gravitational wave interferometers may detect include: a stochastic background level of gravitational waves due to the merging of compact binaries from all over the universe [20], dark matter [21,22], and sources that existed during the early universe [23,24].

To improve statistical models of the systems that emit gravitational waves, the number of detections needs to be increased. To detect continuous wave signals, the signal to noise ratio (SNR) improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The previous, current and future observation runs are summarised in Table 2.1; a more comprehensive overview can be found in [25].

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 30/04/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	2025	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 2.1: Summary of observation runs for LIGO, Virgo and KAGRA [2, 26, 27].

2.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [12] or [13]). The effect these waves have, in the local Lorentz frame, on free test masses on Earth is to move them by a small amount ¹.

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These mirrors are referred to as *end test masses* (ETMs). A suspended mass will behave in the same way as a free mass when it is driven above its resonant frequency, so the ETMs need to be suspended [28]. A

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: All non-gravitational laws of physics which can be expressed in a special relativistic form have the same form in a local Lorentz frame (e.g. [12]). In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 2.1. The amount test masses move, ΔL , when they are at a distance of L away from the origin of the local Lorentz frame due to an incoming plus polarised gravitational wave of strain h is

$$\Delta L = \frac{1}{2}Lh. \quad (2.1)$$

Interferometers are sensitive to arm length changes that are on the order of one wavelength, λ_{EM} , of the light they are operated with. As a plus polarised gravitational wave will move the test masses in anti-phase, the difference between the phase of light when it recombines at the beam splitter after reflecting from the test masses, $\Delta\phi$, is

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}Lh. \quad (2.2)$$

$\Delta\phi$ is known as the *differential phase*, and differential effects result in the light acquiring differential phase. The wavelength current interferometers operate with, $\lambda_{\text{EM}} = 1064 \text{ nm}$, was selected because test masses with low noise and mirror coatings with low loss and low thermal noise can be fabricated for this wavelength [29].

The gravitational wave strain is converted into intensity fluctuations of light at the *antisymmetric* (AS) port due to the two light fields from the arms interfering at the beam splitter (see Figure 2.1). When on the *dark fringe*, there is a small amount of light at the AS port for the signal light to beat with and so the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , in the arm lengths. This technique is known as DC readout

[30, 31]. At the beam splitter, the electric field, Ψ_{AS} , emerging from the AS port due to the interference of the beams returning from the ETMs, Ψ_y and Ψ_x , due to a differential phase, $\Delta\phi = \phi_0 + \frac{4\pi}{\lambda_{\text{EM}}}Lh$, is

$$\Psi_{\text{AS}} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (2.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i\Delta\phi/2}}{\sqrt{2}} - \frac{e^{-i\Delta\phi/2}}{\sqrt{2}} \right), \quad (2.4)$$

where I_0 is the power of the beam injected into the beam splitter by the laser. The intensity of the beam at the AS port, I_{AS} , is

$$I_{\text{AS}} = \Psi_{\text{AS}}^* \Psi_{\text{AS}}, \quad (2.5)$$

$$\simeq \frac{I_0\phi_0^2}{4} + \frac{I_0\phi_0}{2} \frac{4\pi}{\lambda_{\text{EM}}}Lh. \quad (2.6)$$

In the absence of a gravitational wave signal, the DC light level at the AS port is related to the static offset by $\phi_0 = 2\sqrt{\frac{I_{\text{AS}}}{I_0}}$, where $\overline{I_{\text{AS}}}$ is the average of I_{AS} . As we are interested in a fluctuating signal, the DC offset from Equation (2.6) can be removed, allowing for the signal at the AS port to be written as

$$I_{\text{AS}} = \sqrt{I_0\overline{I_{\text{AS}}}} \frac{4\pi}{\lambda_{\text{EM}}}Lh. \quad (2.7)$$

Therefore, it is possible to measure the differential motion of the ETMs due to a gravitational wave's strain by placing a photodiode in the beam emerging from the AS port of a Michelson interferometer.

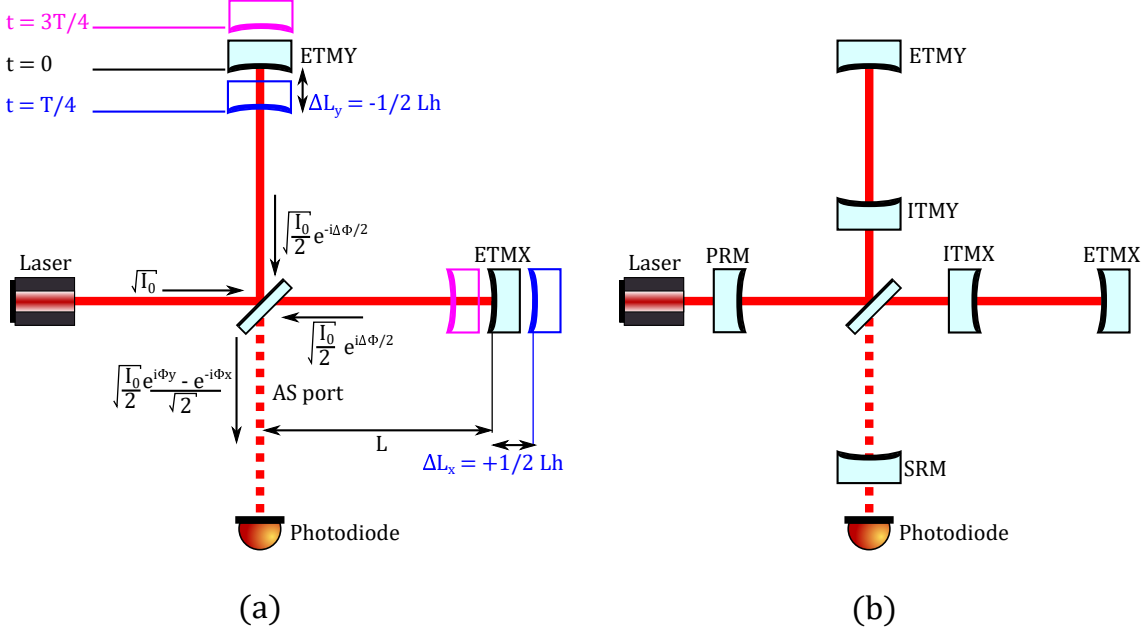


Figure 2.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different times during the gravitational wave's period is shown by the blue and pink mirrors. The electric fields within the interferometer are indicated: $\sqrt{I_0}$ is the laser's amplitude and $\pm\Delta\phi/2$ is the phase acquired by the light as it travels up and down the arms. (b) To reach the sensitivity needed to detect gravitational waves, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

2.3.1 Using Fabry-Perot Cavities in the Arms of a Michelson Interferometer to Enhance Its Sensitivity

The optimal arm length of the interferometer is determined by $1/4$ of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. Gravitational wave signals from BBH and BNS have frequencies of ~ 100 Hz; therefore, the wavelength of the gravitational wave will be of the order ~ 3000 km. It is not possible to construct a Michelson interferometer with ~ 1000 km arms on Earth.

The sites that were available to construct the LIGO detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off of the ETMs, B , needs to be increased. Numerically, B should be ~ 250 . This means that Equation (2.2) becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} BLh. \quad (2.8)$$

Fabry-Perot cavities are used to increase the number of times light bounces off of the ETMs². Cavities are formed by placing an *initial test mass* (ITM) in each arm near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. Interferometers measure the difference in phase of the electromagnetic fields in the arms, $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry-Perot cavity is near or on resonance, the Fabry-Perot cavity enhances the detectors sensitivity as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much

²Herriott delay lines were considered as a means of achieving more bounces from the ETMs, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry-Perot cavities would perform equally well due to the Mizuno limit [32].

greater for a Fabry-Perot cavity than it is for a mirror. This is illustrated in Figure 2.2. The exact way this happens is dependent on the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity acting as a low pass filter (e.g. see [32, Chapter 3], [33, Appendix D], [34] or [35]).

2.4 Sources of Noise in Gravitational Wave Interferometers.

As the strain induced by a gravitational wave causes a minuscule change in the phase accumulated by light in the arm cavities, suppressing the noise below this signal is the main obstacle faced when trying to measure a gravitational wave. From Equation (2.8), for a gravitational wave strain of $h_{\text{rms}} = 10^{-23}$ interacting with LIGO, the beams interfering at the beam splitter will have a difference in phase of $\sim 10^{-11}$ rad.

2.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the interferometer. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The uncertainty in energy, ΔE , that associated with a measurement of the time between two events, ΔT , is determined by the time-energy uncertainty principle, $\Delta E \Delta T \geq \hbar/2$. The uncertainty in the phase, $\Delta\phi$, of a light beam with angular frequency ω_{EM} is related to the measurement time by $\Delta\phi = \omega_{EM} \Delta T$ [36]. The phase is averaged over a time corresponding to half the gravitational wave's period, so $\Delta T = \frac{1}{2f_{\text{GW}}}$. The energy used in the measurement is $E = I_0 \frac{1}{2f_{\text{GW}}} = \hbar \omega_{EM} N$, where I_0 is the power of

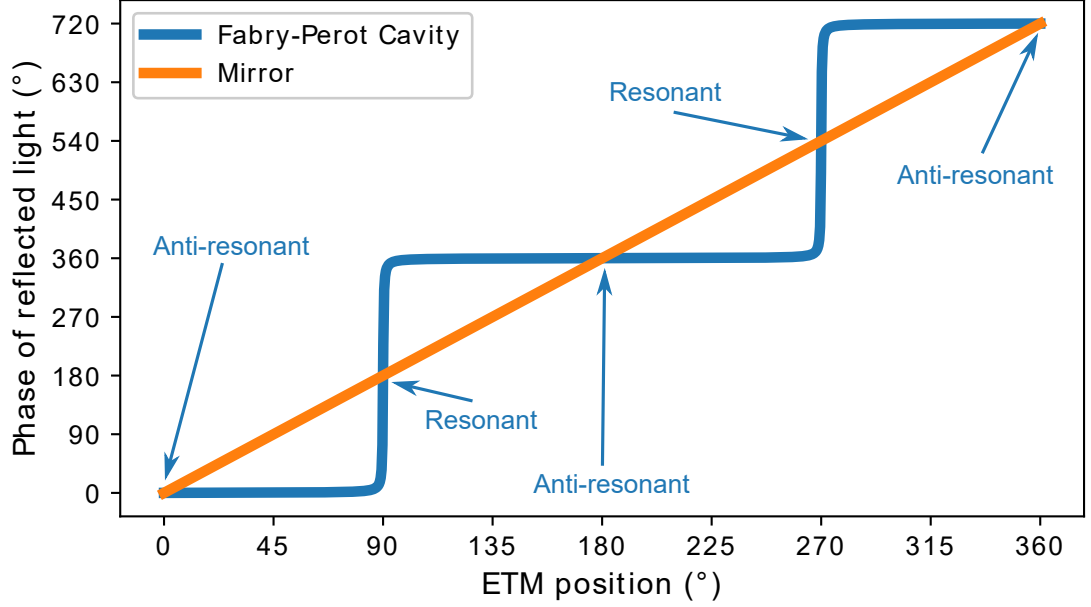


Figure 2.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity as the position of the mirror/end mirror is changed. The orange line shows the extra round trip phase acquired by light in the arm if the ETM is moved one wavelength (360deg) and the blue line shows the same thing except for an Fabry-Perot cavity. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The interferometers sensitivity is proportional to the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$, and so a Fabry-Perot cavity can increase the sensitivity of the interferometer.

the laser and N is the number of photons detected in the averaging time. As the laser is in a coherent state (e.g. [37–39]), the uncertainty in detecting N photons is $\Delta N = \sqrt{N}$; therefore, the uncertainty in the energy used in the measurement is $\Delta E = \hbar\omega_{EM}\sqrt{N}$. Using the expressions for ΔE and $\Delta\phi$, the uncertainty principle may be used to obtain

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (2.9)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the AS port will depend on the amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2\overline{I_{\text{AS}}}\hbar\omega_{EM}\Delta f}. \quad (2.10)$$

The conversion from light power at the AS port to gravitational wave strain can be inferred from Equation (2.7),

$$\frac{h}{I_{\text{AS}}} = \frac{1}{\sqrt{I_0\overline{I_{\text{AS}}}}} \frac{\lambda_{\text{EM}}}{4\pi LB}, \quad (2.11)$$

and so the amplitude spectrum of the strain corresponding to the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c \lambda_{\text{EM}} \Delta f}{4\pi I_0 L^2 B^2}}. \quad (2.12)$$

Equation 2.12 assumed the frequency response of the interferometer to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector’s noise (Figure 2.3).

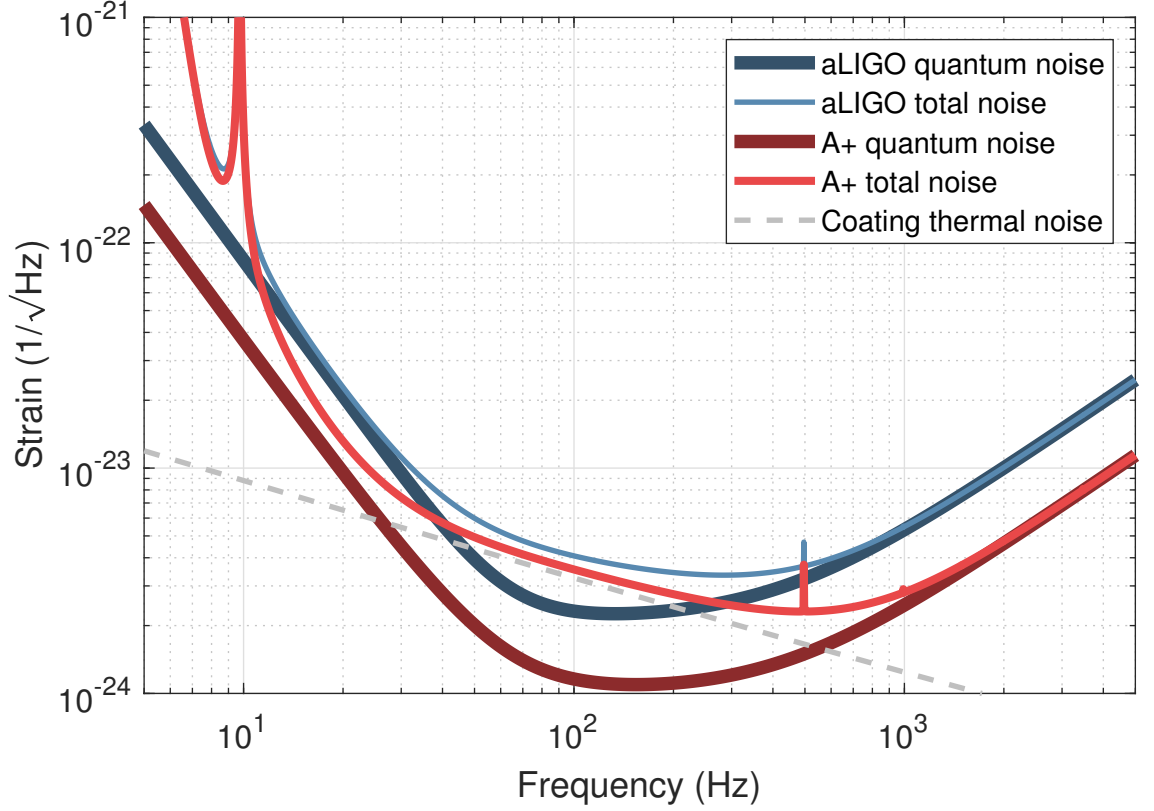


Figure 2.3: The light blue and light red lines show the design sensitivity of the aLIGO and LIGO A+ detectors. The quantum noise, noise due to radiation pressure and photon shot noise, is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise. Below 10 Hz, classical noise sources, e.g. seismic motion, limit the sensitivity of the detector. Above 100 Hz, shot noise limits the sensitivity of both detectors. Shot noise rises with f due to the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure. Coating thermal noise will limit the A+ detectors's sensitivity around ~ 100 Hz.

To beat the photon shot noise by a factor of 10, Equation (2.8) and Equation (2.9) show that, for a LIGO-like interferometer, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave signal with a frequency of 100 Hz, the laser power incident on the beam splitter would need to be $\sim 1\text{MW}$. No such laser exists.

2.4.2 Using Power Recycling to Reduce Shot Noise

To increase the laser power that is incident upon the beam splitter, a power recycling cavity can be used [40–42]. When the interferometer is on the dark fringe, most of the light from the laser gets reflected by it. Thus, the interferometer is, effectively, a mirror. Drever [43] [can't find a copy!](#) proposed that a *power recycling mirror* (PRM) could be placed between the laser and the interferometer to form a cavity. A sketch of this is shown in Figure 2.1. The total power impinging on the beam splitter would be increased by making wavelength-scale position changes, i.e. *tuning*, of the position of the PRM such that the light reflected from the interferometer and the light from the laser add in phase, i.e. the reflected light power would be ‘recycled’. Current laser sources that can be operated for year-long timescales are capable of producing of the order 100W, see Chapter 3, and so the power recycling factor needs to be $\sim 10^4$. This technique has been implemented at LIGO [44] and other large-scale interferometers [26, 27].

2.4.3 Using Signal Recycling to Shape Quantum noise

An interferometer’s bandwidth depends on the amount of time that signal-carrying light is in the arms for. Meers suggested that the signal light could be recycled in

a similar fashion to how power is recycled [40]. This detection bandwidth can be altered by introducing a *signal recycling mirror* (SRM). A coupled cavity, known as the *signal recycling cavity* (SRC), between the SRM, ITM and ETM will be formed by placing the SRM between the photodiode and the beam splitter. A sketch of this is shown in Figure 2.1. The finesse of the arm cavities as seen by the light emerging at the AS port depends on the tuning of the SRM. There are two common modes of operation: *resonant sideband extraction* (RSE) [45,46] and *signal recycling*. Alternatively, the SRM can be tuned so that the interferometer is optimised for a narrow band of frequencies, e.g. [47].

When in RSE mode the SRM is tuned so that the SRC is anti-resonant, thus broadening the interferometers sensitivity bucket at the expense of losing peak sensitivity. This was motivated by needing to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities [46]. However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To retain the desired arm cavity finesse from a sensitivity point of view, the SRM can be tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that the SRC is resonant. This deepens the sensitivity bucket at the expense of bandwidth. The trade-off between bandwidth and peak sensitivity comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth; so for the same amount of power at the beam splitter, this can be thought of as a fundamental limit to the interferometer's sensitivity. This is known as the Mizuno limit [32].

2.4.4 Reducing Shot Noise by Removing Junk Light with a Mode Cleaner Cavity.

All light that is incident on the output photodiode contributes to the total shot noise of the measurement; therefore, only light which carries signal should be incident on the photodiode to enhance the SNR of the measurement. In an ideal interferometer, the light emerging at the AS port will be a beam that is defined by the mode that is resonant within the arm cavity. To minimise diffraction losses and due to the mirrors's manufacturing tolerances, the light in the arm cavity will be in the *fundamental Gaussian mode*. However, due to optical imperfections, e.g. astigmatism, and the RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*.

To remove junk light, mode cleaners are used (e.g. [48, 49]). A mode cleaner is a high finesse cavity that strips away higher order mode light and the control sidebands. As higher order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity's finesse. However, it is possible for some higher order modes to not be suppressed if their round trip Gouy phase is nearly the same as that of the fundamental mode; thus, the cavity is designed so that only higher order modes which carry an insignificant amount of light will do this. The control sidebands are attenuated if they are outwith the cavities linewidth. The beam transmitted by the mode cleaner will be

predominantly in the fundamental mode, and so the SNR of the signal measured by the photodiode will be increased.

2.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test masses when it is reflected from them [37]. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the positions of the ETMs. As the force exerted by a beam is $F = \hbar\omega_{EM}N/c = P/c$, the amplitude spectral density due to the radiation pressure of the length of a Fabry-Perot cavity with mirrors of mass m is

$$\Delta x_{\text{RPNFP}} = \frac{\sqrt{\hbar\omega_{EM}P}}{2mc\pi^2 f^2}. \quad (2.13)$$

To reduce the effect of radiation pressure noise, the LIGO test masses are 40 kg. The effect of radiation pressure noise on the LIGO detectors's sensitivity curves can be seen in Figure 2.3.

Radiation pressure noise scales with the quantum fluctuations entering the AS port, whereas shot noise is inversely proportional to these fluctuations. For a laser which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [37, 50, 51] sets an optimal laser power for a given ETM mass which minimises the total quantum noise. The SQL results in a $1/f$ locus that is determined by the mass of the ETM and the duration of the measurement.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [52].

2.4.6 Coating Thermal Noise and Other Noise Sources

There are many classical sources of noise to overcome in a gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of atoms in the amorphous mirror coating layers of the test masses [53]. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector, as shown in Figure 2.3. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in the Earth’s crust, residual gas pressure noise that alters the refractive index of the arms, and the noise associated with the interferometer’s electronics (e.g. [54, 55]).

2.5 Reducing Shot Noise with Non-Classical States of Light

The two-photon formalism developed by Caves and Schumaker [56, 57] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO). A SHO can be described by two orthogonal components, such as the momentum and

position quadrature; for light a convenient basis to use is the *amplitude* and *phase* quadrature. The amplitude and phase quadratures, $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively, defined in terms of annihilation and creation operators of the electromagnetic field, $\hat{a}$ and $\hat{a}^\dagger$, are

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) , \quad (2.14)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) . \quad (2.15)$$

From this formalism, an uncertainty relation between the phase and amplitude quadratures can be derived:

$$\Delta\hat{\mathbf{X}}\Delta\hat{\mathbf{Y}} \geq \frac{1}{2} \quad (2.16)$$

If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signals will be in the phase quadrature of the light, it would be advantageous to, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature; this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting single higher energy photons into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at the LIGO Livingston Observatory (LLO) [58] and GEO600 [59].

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the AS port. During LIGO’s third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise from limiting the BNS range of the detector [58].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field [60]. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the combined beam’s intensity. The relative phase between the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter 5.1 explores this in more detail.

2.6 Interferometry Techniques to Beat Quantum Limits

Squeezed light and an effect known as ponderomotive squeezing [50], can be used to surpass the SQL at some frequencies. Ponderomotive squeezing arises due to the natural correlations between the phase and amplitude quadratures of the light created by the interferometer. A demonstration of the SQL being surpassed at the LIGO Livingston detector can be found in [61]. Although [61] shows the SQL was beaten at some frequencies, the noise at other frequencies was increased relative to if no squeezing was used. Although the LIGO detectors are operated for broadband gravitational wave sensitivity, sub-SQL interferometer configurations could be used if detecting certain astrophysical sources with merging frequencies around the

frequency at which the SQL is beaten was of interest.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency dependent squeezer can be utilised [50]. The light's squeezing angle can be rotated as a function of frequency. This is done by using a high finesse detuned filter cavity. At frequencies below the linewidth of the cavity, the light reflected from the filter cavity will experience a phase shift and so the squeezing angle will be rotated; in contrast, at frequencies above the cavity's linewidth, the light incident upon the cavity is reflected without experiencing any phase shift and so the squeezing angle is not rotated. At low frequencies, the amplitude quadrature can be squeezed and thus the interferometer's radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and thus the interferometer's shot noise can be decreased. A frequency dependent squeezed light source that is suitable for LIGO has been demonstrated [62] and frequency dependent squeezing will be part of the A+ detector.

Variational readout is a technique where quantum noise due to radiation pressure is removed from the measurement by using a frequency dependent homodyne angle [50]. This is realised in a similar way to frequency dependent squeezing; the AS beam is coupled to a filter cavity and thus the detected quadrature of the beam emerging from this cavity is frequency dependent. This beam is then measured on photodiodes using a conventional BHD scheme. Variational readout schemes are sensitive to small amounts of loss within the filter cavity, and so this technology is not mature enough to be implemented into current gravitational wave detectors.

To beat the bandwidth limit, energy has to be injected into the system. White-light cavities as a means of increasing the detector bandwidth have been studied [63–65]. However, these interferometer configurations have yet to be experimentally demonstrated.

There are several sub-SQL schemes which avoid the quantum noise problems related to making a position measurement of the test masses. If a momentum measurement of the test masses could be made, then a random impartation of ‘position’ would not affect future measurements of the momentum [66]. Such measurements are known as quantum non-demolition. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$; this type of measurement can be made with a speed meter interferometer [67–69].

2.7 The Effect Losses Have on Measuring Quantum States of Light

To preserve the benefits gained by using quantum states of light, it is essential to have minimal losses in the readout of a gravitational wave detector. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss (e.g. [38, Section 2]). The incoming light enters a beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal component, and so the

state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss. The factor by which the shot noise increases, ϵ , due to a simple loss mechanism, such as a non-ideal photodiode (see Section 2.8), is given by

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}, \quad (2.17)$$

where $0 < \eta < 1$ is the loss.

Mode matching losses have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism [71]. This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss. This is the focus of Chapter 6.

2.8 Photodiodes

Photodiodes are used to sense the light emerging from the OMC. A photodiode is a square law detector that produces a current that is proportional to the power of the light it is detecting. As photodiodes are low-passed square law devices they produce a signal with a DC component and components proportional to the frequency and amplitude modulations of the field [129]. Usually, photodiodes are of the order 1 mm in diameter, have a relatively high quantum efficiency and a fast response. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) semiconductor structure. PIN photodiodes can typically detect up to 100 mW of light power.

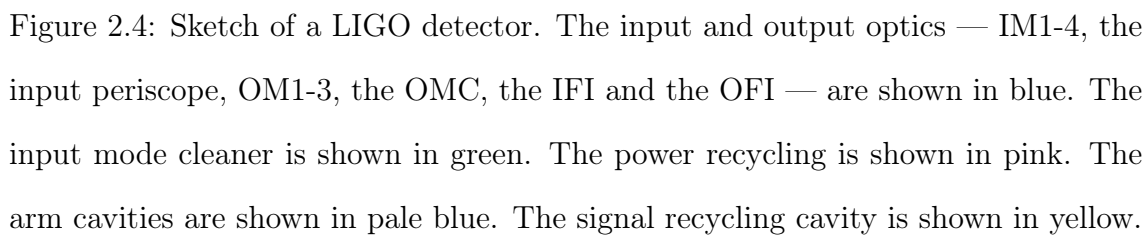
Photodiodes convert photons into electrons, ideally with a one-to-one ratio. The measure of how well this is done is the photodiode’s *quantum efficiency*. Due to the size of its bandgap, InGaAs is a widely used semiconductor for detecting 1-1.6 μm light. Suitable InGaAs photodiodes were available for detecting the signal-carrying light in the current generation of gravitational wave detectors; the requirements for these photodiodes is discussed in Section ??.

Since future gravitational wave detectors (see Section 2.10) may be operated with 2 μm light, the current photodiodes may be replaced. ‘Extended’ InGaAs photodiodes may be used instead. These photodiodes are extended in the sense that their cut-off wavelengths are above 1.6 μm . However, at the time of writing (2020) and to the author’s knowledge, commercially available InGaAs photodiodes for detecting 2 μm light that are suitable for use as the readout photodiode in a gravitational wave detector do not exist [110–113]. To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium compared to conventional InGaAs photodiodes; however, this extra indium causes a lattice mismatch between the InGaAs and the InP substrate (see e.g. [122]). As discussed in Chapter 8, the defects due to the strain resulting from by the lattice mismatch can cause the noise and quantum efficiency of the photodiode to be degraded. Previous experiments in squeezing at 2 μm have been limited by the properties of the extended InGaAs photodiode that was used [123, 124]. To obtain low enough electronic noise in [123], the photodiodes were not biased; this came at the expense of the photodiode’s quantum efficiency and thus the total level of observed squeezing was reduced. Chapter 8 contains the characterisation of some extended InGaAs photodiodes.

2.9 Overview of the LIGO Detector

The LIGO detector is a dual-recycled-Fabry-Perot Michelson interferometer with 4 km arms [2]. The layout of the detector is shown in Figure 2.4 [72] and its sensitivity curve is shown in Figure 2.3 [73, 74]. The term dual-recycled refers to the fact that the interferometer uses both power and, the mathematical dual of power recycling, signal recycling. The interferometer is housed within a vacuum system comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 2.4 includes all the optics from the pre-stabilised-laser (PSL) to the readout photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors, known as PR2 and PR3, in addition to the PRM. The role of these mirrors is to make the round trip Gouy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason [44]. The beam passes through the output Faraday isolator (OFI), and is steered into the output mode cleaner. The gravitational wave signal is encoded in the beam transmitted by the OMC, so photodiodes monitor this light.



2.10 The Future of Ground Based Gravitational Wave Detectors

The next generation of ground-based gravitational wave detectors include the Einstein Telescope [75,115], LIGO Voyager [76] and LIGO Cosmic Explorer [77,109,116]. LIGO Voyager will be a cryogenic interferometer installed in the current LIGO infrastructure and will be a factor of ~ 2 more sensitive than LIGO A+. The Einstein telescope and Cosmic explorer interferometers will be ten times larger than the current generation of interferometers and will be able to: test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic sources which have not yet been theorised.

The current generation of gravitational wave detectors will be limited by the thermal noise around 100 Hz, see Figure ??, due to the Brownian motion of the molecules that make up the mirror coatings [114]. Since these thermally driven fluctuations are a limiting source of noise, proposed future gravitational wave detectors, such as the Einstein Telescope [75,115] and LIGO Cosmic Explorer [116], may use cryogenically cooled mirrors.

However, the material that the test masses are made from, fused silica, has undesirable noise properties when it is cooled [120] and so the test masses may be made

of crystalline silicon; crystalline silicon is opaque at $1\text{ }\mu\text{m}$ [121] and so this also forces a change from a $1\text{ }\mu\text{m}$ laser. Additionally, cryogenically cooled mirror coatings may not be compatible with the current laser wavelength [117–119], which is $1\text{ }\mu\text{m}$, and so future gravitational wave detectors may use either a $1.5\text{ }\mu\text{m}$ or $2\text{ }\mu\text{m}$ wavelength laser [77, 109]. $2\text{ }\mu\text{m}$ is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength [78].

2.11 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. For this to be successfully implemented, the quantum state of the light field must be accurately detected with a balanced homodyne readout and losses in the readout must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with $2\text{ }\mu\text{m}$ light, future detectors may use a $2\text{ }\mu\text{m}$ laser; this leads to the question: is laser technology at this wavelength sufficiently mature to enable quantum limited measurements? Balanced homodyne detection and photodiodes with high quantum efficiency and low dark noise will be required for any future interferometer.

Chapter 3

A Prototype High Power Laser for LIGO's Fourth Observation Run

Chapter 4

The Consequence of Radius of
Curvature Errors within the SRC on
Modes at the SRM.

Chapter 5

Balanced Homodyne Detection for LIGO A+

note: check the main reason for BHD, cite Eric and Lee. A major change between aLIGO and LIGO A+ will be the replacement of DC readout with balanced homodyne detection (BHD). BHD plays a key role in many experiments that utilise non-classical states of light (for a recent example of such an experiment relating to gravitational wave interferometry, see [61]). The transition from aLIGO to LIGO A+ involves the redesign of many aspects of the interferometer, and elements of this effort aim to exploit quantum effects in order to increase the signal to noise ratio (SNR) of the detector. Therefore, the implementation of BHD is an attractive prospect for those who want to harness the quantum nature of light to enhance the detector.

This Chapter begins by exploring the fundamentals of BHD. Section 5.1 contains a description of BHD and how it will benefit LIGO. For the implementation of BHD to be successful, the phase noise between the local oscillator and the signal beam needs to be low enough as to not degrade the sensitivity of the detectors. BHD may introduce phase noise between the LO and signal. Section 5.2 derives the phase noise requirement for the LO beam for A+. Section 5.3 concludes this Chapter.

5.1 What is Balanced Homodyne Detection?

BHD is a method of measuring information encoded by a light field, and it forms a key part of many quantum optics experiments. By combining a signal beam with a much stronger LO beam, phase modulations in the signal beam become amplitude modulations in the combined LO-signal beam (e.g. [38]). BHD can be configured to measure any quadrature of the signal beam which is vital for inferring the quantum state of a system and is required for more complex topologies of interferometer [50,68,79,80]. BHD has been implemented to detect 11.6 dB of shot noise suppression to below 1 Hz due to the squeezing of vacuum fluctuations. [81].

All readout schemes for gravitational wave detectors rely on a LO field of some kind. The first generation of detectors used heterodyne readout (e.g. [82]). Heterodyne readout involves a modulation-demodulation scheme whereby the signal is mixed with a RF LO to a frequency where technical noise is no longer a concern. It is then demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the quantum shot noise is enhanced by the modulation-demodulation process [83].

The LO in a BHD readout scheme is a light field with the same optical frequency as the signal beam. The LO beam has a higher power than the signal beam and acts as a stable reference frequency. Ideally the LO would have no phase noise, however in reality there are fluctuations in the phase between the LO and the signal beam. The interference of the LO and signal beam causes this phase noise to become an amplitude modulation in the combined LO-signal beam [84]. This phase noise must be kept below the size of the phase fluctuations in the signal beam that are to be measured, which in A+ are determined by the shot noise of the light within the arm cavities. Unlike Heterodyne readout, BHD does not cause a fundamental increase in the noise floor of the measurement.

BHD is implemented by injecting the LO and signal beams into different input ports of a balanced beam splitter and placing photodiodes in the remaining output ports. The information encoded in an arbitrary quadrature of the signal beam can be selected by correctly setting the relative phase between the LO and signal beams. The relative phase between the LO and signal beam is also known as the *homodyne angle*. Subtracting the two photo-currents results in a measurement which is proportional to the selected quadrature of the signal beam. A simple BHD layout is shown in Figure 5.1.

Due to the balanced nature of the beam photocurrent, the classical intensity noise of the LO is not present in the BHD signal. This and the ability to set the homodyne angle make BHD ideal for measuring quantum states of light. For example, BHD can be used to detect the level of squeezing in a quadrature of the signal beam by setting the homodyne angle to the quadratures angle (e.g. [60]). From the two-

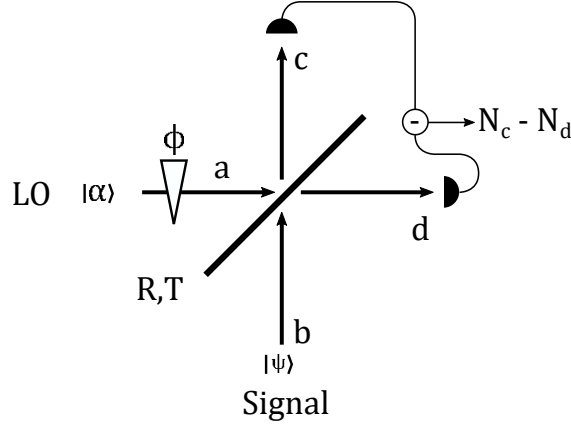


Figure 5.1: A coherent state $|\alpha\rangle$, enters port a of a beam splitter. A signal $|\Psi\rangle$ enters port b . These two beams interfere, resulting in photons being detected at the c and d ports. The two photo-currents are subtracted, and the signal that is produced is proportional to an arbitrary quadrature of the state $|\Psi\rangle$. The quadrature that is measured is determined by the homodyne angle, ϕ . This is expressed mathematically by Equation (5.1).

photon formalism of light (see Section 2.5), the general quadrature operator, $\hat{\mathbf{X}}_\phi$ can be obtained,

$$\hat{\mathbf{X}}_\phi = \hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi, \quad (5.1)$$

where ϕ is the homodyne angle. By setting the homodyne angle, one can sense an arbitrary combination of the phase and amplitude of a field.

Consider a LO field $\hat{a}$ and a signal field $\hat{b}$. It is arranged so that these fields interfere on a beam splitter and are detected on two photodiodes, c and d . The

fields these diodes detect are $\hat{c}$ and $\hat{d}$. The photo-currents are then

$$\hat{\mathbf{N}}_c = \hat{c}^\dagger \hat{c} = \left(\sqrt{R} \hat{a}^\dagger + \sqrt{T} \hat{b}^\dagger \right) \left(\sqrt{R} \hat{a} + \sqrt{T} \hat{b} \right) \quad (5.2)$$

$$= R \hat{a}^\dagger \hat{a} + T \hat{b}^\dagger \hat{b} + \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \quad (5.3)$$

$$\hat{\mathbf{N}}_d = \hat{d}^\dagger \hat{d} = T \hat{a}^\dagger \hat{a} + R \hat{b}^\dagger \hat{b} - \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (5.4)$$

For BHD, the beam splitter must be balanced so $R = T = 1/2$. This results in the difference in the photo-currents being

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = \hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}. \quad (5.5)$$

The operators $\hat{a}$ and $\hat{b}$ can be ‘linearised’, $\hat{a} \rightarrow \alpha + \delta a$ in order to separate the quantum behaviour and classical behaviour. Furthermore, the phase between the signal field and the LO field can be expressed in terms of the homodyne angle with $\hat{b} \rightarrow \hat{b} \exp(i\phi)$. By neglecting terms of the order $\delta a \delta b$, this allows one to rewrite equation (5.5) as

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b + 2\beta\delta \mathbf{X}_\phi^a. \quad (5.6)$$

The variance, $\sqrt{V(X)} = \Delta X$, of this measurement would then be

$$V(\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d) = 4\alpha^2 V(\delta \mathbf{X}_{-\phi}^b) + 4\beta^2 V(\delta \mathbf{X}_\phi^a). \quad (5.7)$$

If the LO is much stronger than the signal field, $\alpha \gg \beta$, then the measurement is

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b, \quad (5.8)$$

which is proportional to the general quadrature of the signal.

5.1.1 Motivation for Implementing BHD in LIGO A+

Without BHD, the LIGO detectors will not be able to reach the design sensitivity for A+. To reach the level of frequency dependent squeezing required to meet the A+ design sensitivity, the quadrature angle of the light must be a free parameter. Currently, DC readout is used [31]. With DC readout, the homodyne angle can be set; however, practically it is constrained as this requires introducing an imbalance of power between the two arms [85]. There is a low amount of tolerance on the homodyne angle before the quantum noise is significantly degraded []; thus, the homodyne angle must be a free parameter. Complete freedom in the homodyne angle can be achieved with BHD.

Different gravitational wave sources carry their information in different frequency bands and the detector may be optimised to reflect this [50, 71, 86, 87]. For instance, BNS signals continue to higher frequencies than BBH signals so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS systems. To optimise the sensitivity of the interferometer to different sources the detectors noise would need shaping, a task that can be achieved, along with other upgrades (e.g [19, 50]), BHD.

A squeezed light field can be used to surpass the standard quantum limit (see Chapter 2), and to detect these squeezed states the homodyne angle must be correctly set. The exact way in which the quantum noise is changed is controlled by the squeezing angle. Currently the squeezing angle is set to be at the same angle

as the readout quadrature, rather than the preferred option of setting the readout quadrature to the squeezing angle. The effect that controlling the squeezing angle has on the quantum noise floor has been demonstrated at LLO [61]. However, [61] used a diagnostic BHD set-up which was not intended for providing the low noise required to make gravitational wave detections.

More advanced upgrades (see Chapter 2.6 or [50]) could alter quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are non-trivial, and would require significant redesign of many parts of the layout, so the implementation of BHD for A+ does not necessarily correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

BHD has some practical benefits over DC readout [85]. The practical benefits arise due to being able to eliminate the DC light required for DC readout to work. One benefit is in the reduction in SRC length noise coupling to the gravitational wave signal¹. Another benefit is being able to reduce the scattering from the septum

¹The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity. For small changes in cavity length, the light transmitted by a cavity that is on resonance or anti-resonance will be modulated by a term proportional to the square of the length change. This signal due to the length of the SRC is noise. The amplitude of this noise is linearly dependent on the amount of light entering the cavity which is a result of the DC offset. By decreasing the DC offset, this noise is decreased. As this noise is in the amplitude quadrature of the signal beam, unlike the gravitational wave signal which is in the phase quadrature, it will not be amplified by the LO if the homodyne angle is set to readout from the phase quadrature. The second route SRC length noise may couple to the signal is via the optical springs between the SRC and arm cavities. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity lengths to change as well [88].

window between HAM5 and HAM6.

5.2 Local Oscillator Path Length Stability Requirement for A+

In this section, a semi-classical approach is used to derive the path length stability requirement for the local oscillator. The end result is the same as if a full quantum mechanical treatment was used (e.g. [84]). The quantum mechanics of this problem are included in the A+ sensitivity curve and the expected power in the signal beam. The phase noise of the LO must be kept low enough so that it does not result in intensity fluctuations greater than those due to the shot noise of the light in the signal beam. This requirement was also derived in [89], [90] and [91]; however, in this section, the most current layout was considered.

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable from phase fluctuations in the signal beam. A gravitational wave produces a phase modulated signal in the signal beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be low enough for the signal that is generated by the interference of the LO and signal beams to be due to the phase modulation of the signal beam.

The relationship between the differential phase in the arms and the power in the signal beam was used to find the power noise for the beam exiting the SRM in A+. The linear response function due to a differential phase change in the arms was

computed with FINESSE. This was used to deconvolve the strain sensitivity curve to produce a power noise curve for the light exiting the AS port of the interferometer. Since the interferometer is linear, noise is transformed linearly. This means that the quantum nature of the noise is irrelevant to the calculation.

This light noise curve was then used to set the maximum allowable intensity fluctuations due to phase noise between the LO and the signal beams. This phase noise can then be converted into a distance, and this produces a stability requirement for the steering optics used to combine the LO and signal beams.

The aforementioned steps will be gone through in the following sections. First, the LO path will be described. Then a linear response function between a differential strain and signal at the AS port will be obtained via FINESSE. Next, the sensitivity curve for A+ [74] will be expressed in units of $\frac{W}{\sqrt{Hz}}$. This will be converted into a longitudinal motion requirement. Finally, there is a short discussion about the proposed optics that fulfil this requirement.

5.2.1 Local Oscillator Path

The beam path for the LO and signal beams up to the point of recombination is described in this section. The layout for the optics which combine the LO and signal beam is shown in Figure 5.2. The LO will be picked off from the power recycling cavity and guided by one steering mirror, BHDM1, to the BHD combining beam splitter, BHDBS2. The signal beam will exit the OFI and be steered by one mirror, OM0, onto BHDBS2. OM0 and BHDM1 will be used to align the beams so they overlap on BHDBS2. The angle of BHDBS2 will be used to make the LO and signal

beams overlap in the far field. Two lenses are used for mode matching the LO beam to the signal beam.

The LO could have been derived from a separate laser, but to acquire laser light with low frequency and intensity noise the LO will be picked off from a point within the interferometer. Two options for the LO pick off were considered [89]: a beam picked off from the main beam splitters AR coating (BSAR), or the beam transmitted through PR2 (POP). The determining factor in choice of pick off point was the effect of backscatter. The POP port was chosen as it is far superior to the BSAR choice at all frequencies and scatter phases. This is because backscattered light from the BSAR enters only one arm, thus making it a differential affect. Any backscatter that enters the POP port is common-mode, and therefore DARM is far less sensitive to this source of backscatter.

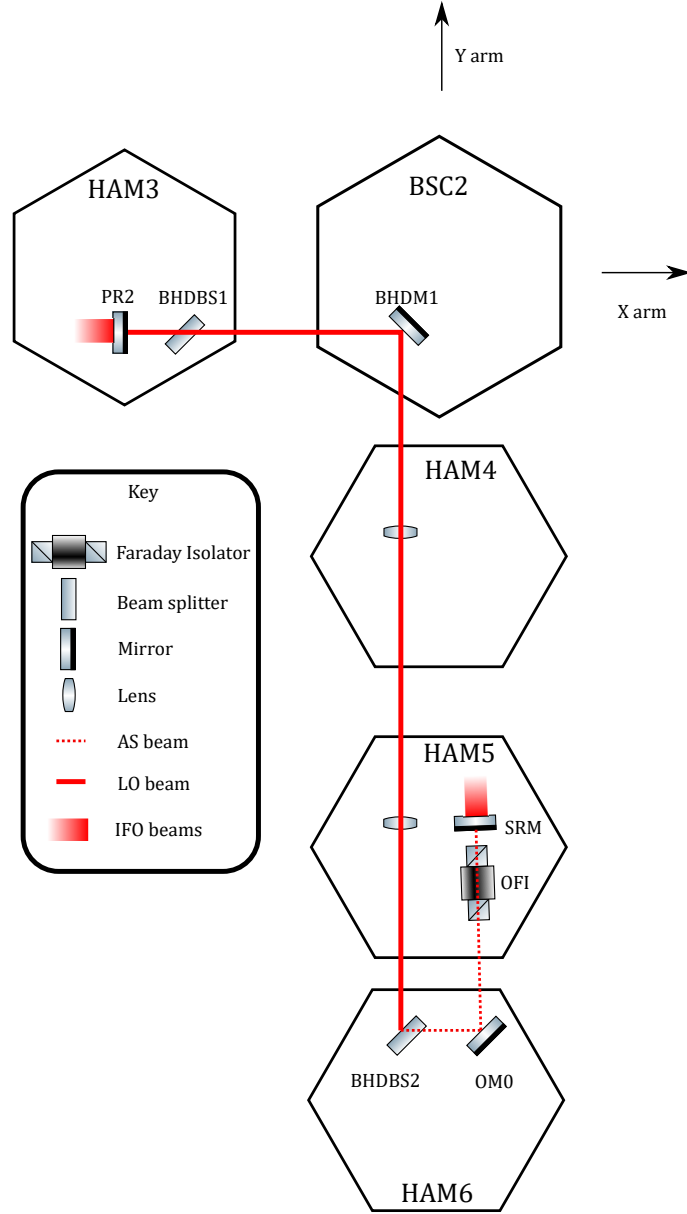


Figure 5.2: Sketch of the LO and signal beam paths in relation to other parts of the interferometer, and the HAM tables are shown by the hexagons (see Figure 2.4) Section ??). Most of the interferometer has not been shown for clarity. The main interferometer beam is indicated by a thick, gradated red line. The thinner red line indicates the LO beam, and the signal beam is shown with a dashed red line.

5.2.2 The Linear Response Function Between Differential Strain and Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model [92] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. The linear response function at the frequencies contained in the detection band are not influenced by the OMC. This means no major modifications to the FINESSE model [92] were needed, and so the model represents a state that the interferometer would be in for observing gravitational waves.

FINESSE allows one to simulate linear response functions by applying a ‘swept sine’ signal to a component and demodulating the light at the frequency of the applied signal. The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency.

For this calculation, the linear response function measured in FINESSE needs to be converted into more convenient units. To get the linear response function between arm motion and AS power into units of $\frac{\text{Power}}{\text{Strain}}$, or $\frac{\text{W}}{\text{h}}$, several factors are required. Since the input modulation is doubled by moving the mirrors differentially, the response needs dividing by two. To account for the ETMs retro-reflecting the light, the response needs dividing by two again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in Figure

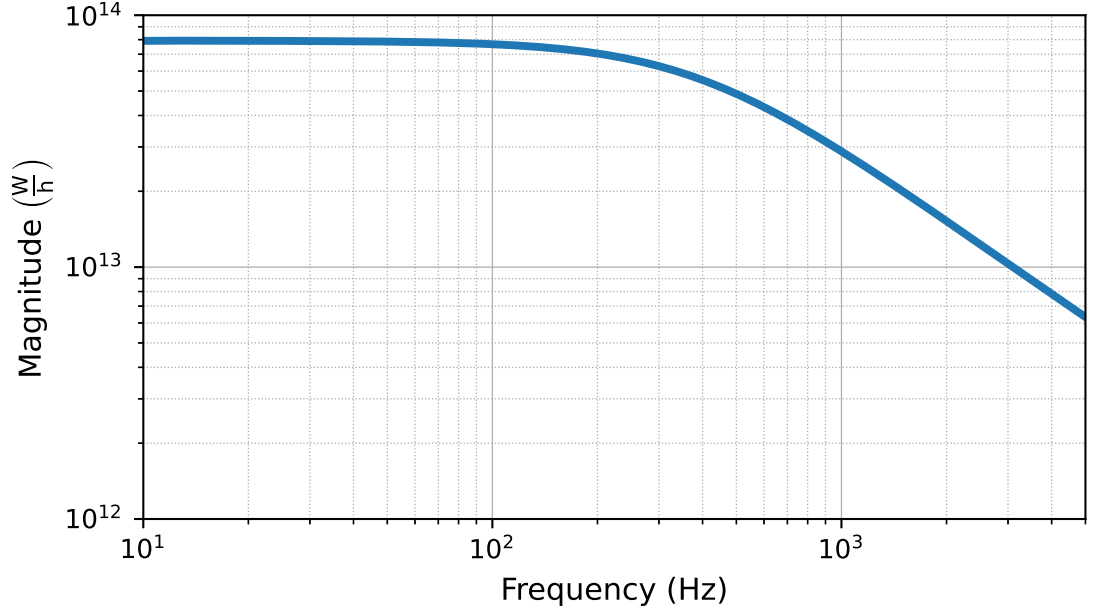


Figure 5.3: The linear response function between strain and light power, expressed in units of Light Power/gravitational wave strain, at the AS port was obtained from a FINESSE simulation of the LLO detector. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity.

5.3.

5.2.3 Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement

To convert the A+ sensitivity curve into a path length stability requirement for the LO, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{\text{W}}{\sqrt{\text{Hz}}}$. Second, the intensity noise of the signal beam needs to be converted into a phase noise. In order for the phase noise of the LO to not limit the performance of the gravitational wave detector, the phase signal in the signal beam must be

greater than the phase noise of the LO beam. Finally, a factor of ten safety margin is required. The HAM Relay Triple Suspension (HRTS) [93] requirement is then obtained by including the number of reflective optics in the LO path and signal path.

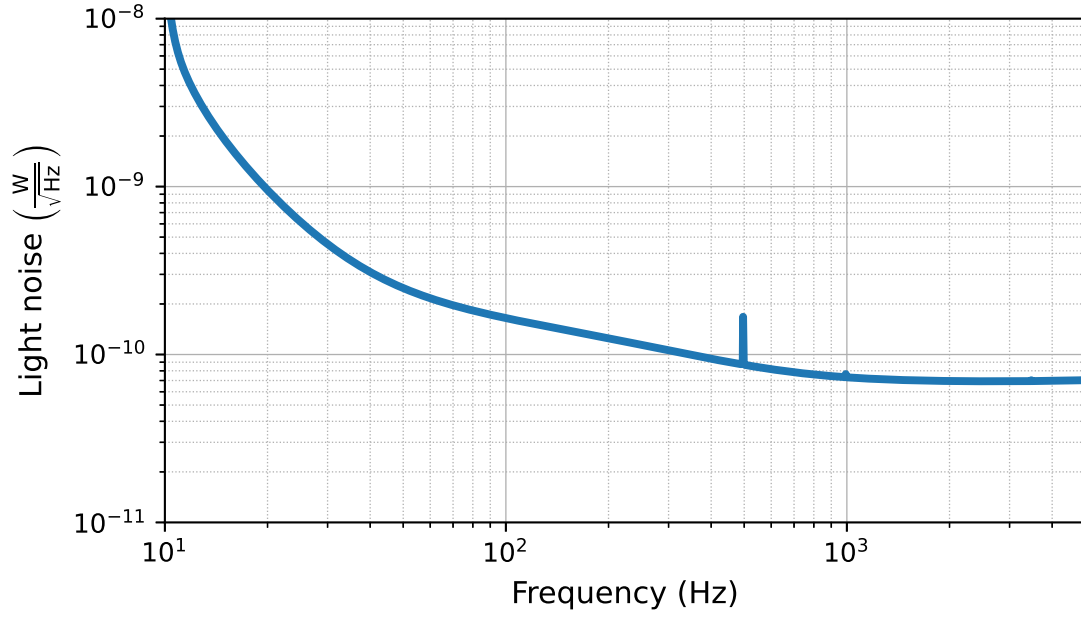


Figure 5.4: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve with the function shown in Figure 5.3. At frequencies above 1 kHz the noise is flat as it is dominated by shot noise. At lower frequencies the noise is increased relative to shot noise due to radiation pressure, coating thermal noise and other noise sources. The spike in the spectrum at 500Hz represents the presence of suspension modes (actually find out what this is).

The A+ sensitivity curve was multiplied by the function shown in 5.3 to convert it from units of $\frac{\text{h}}{\sqrt{\text{Hz}}}$ to $\frac{\text{W}}{\sqrt{\text{Hz}}}$. This is shown in Figure 5.4. At frequencies above 1 kHz the requirement is flat as this is the region dominated by shot noise. At lower frequencies the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

The change in power measured in the combined LO/signal beam due to a phase modulation, i.e. path length modulation, in the LO or signal path needs to be calculated. When the LO and signal beams interfere at BHSBS2, the exiting beams have power P_c and P_d . These are given by

$$P_c = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} + \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} + |E_s E_{\text{LO}}^\dagger| \cos(\phi) \quad (5.9)$$

and

$$P_d = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} - \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} - |E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (5.10)$$

where E_s and E_{LO} are the (complex) field amplitude of the signal and LO beams; ω is the laser frequency; t is time and ϕ is the homodyne angle. The difference in these signals gives BHD signal

$$P = 2|E_s E_{\text{LO}}^\dagger| \cos(\phi). \quad (5.11)$$

Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{\text{LO}}^\dagger| \sin(\phi) d\phi, \quad (5.12)$$

and since the homodyne angle is 90° this becomes

$$dP = -2|E_s E_{\text{LO}}^\dagger| d\phi. \quad (5.13)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi} d\phi = dx. \quad (5.14)$$

The light noise curve shown in Figure 5.4 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the noise floor of the interferometer in units of LO path length noise.

To determine the longitudinal motion requirement for the optics that combine the LO and signal beam the number of optics and, since there are many uncorrelated noise sources in LIGO, a factor of ten safety margin needs to be included. The requirement assumes, based on the current design, that there will be two mirrors at 45° which are used to steer the AS and LO beams. As the optics reflect at 45° , the path length change due to the optics motion is multiplied by $\sqrt{2}$. The beams are combined on a beam splitter, and the motion of this beam splitter needs to be counted twice. Figure 5.5 illustrates this.

For a 98 mW LO beam and a 5 mW signal beam, the noise floor of the interferometer in units of LO path length is shown in Figure 5.6. The requirement for the longitudinal motion of optics in the LO or signal path at 10 Hz is $5 \times 10^{-13} \text{m}/\sqrt{\text{Hz}}$.

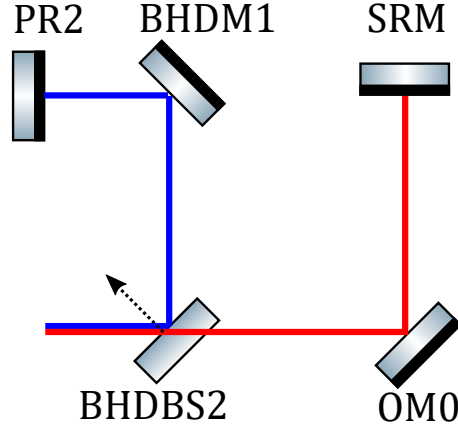


Figure 5.5: A sketch of the LO and signal beam paths. The LO beam path is shown in blue, and the signal beam is shown in red. A dashed arrow indicates the direction that the recombining beam splitter, BHDBS2, may move in. If BHDBS2 moves in this way, the phase of the signal beam has increased while the phase of the LO beam has decreased at the point where the signal and LO beams combine. This means that the recombination beam splitter motion needs to be counted twice in calculations of the phase noise requirement.

To determine the amount of isolation required, the typical motion of the optical table needs to be compared to the longitudinal motion requirement for the optics. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-9} \text{m}/\sqrt{\text{Hz}}$ [94]. To achieve this with passive isolation, triple suspensions will be used.

5.3 Conclusion

One of the major changes to LIGO for A+ will be the implementation of BHD. BHD is essential to ensure that the LIGO detectors will meet the sensitivity requirement for A+ as frequency dependent squeezing requires the homodyne angle

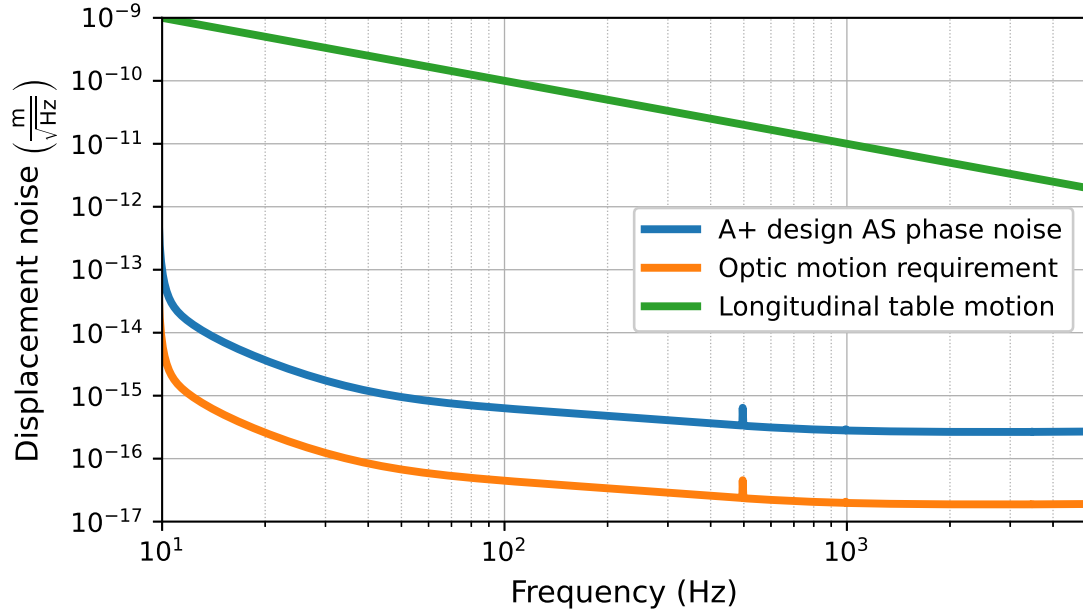


Figure 5.6: The phase noise between the LO and the signal beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion. The table motion was modelled as $1/f$ noise.

to be a free parameter.

Details of the LO path were discussed and analysed. The LO will be derived from a beam picked off from the POP port, and a sketch of how the signal and LO will be combined is shown in Figure 5.2. Starting with the linear response function between a differential strain and a signal at the AS port and the strain sensitivity curve for A+, we were able to derive a relative path length stability requirement for the LO and signal beams. By comparing this motion to a typical motion of the optical table, we were able to derive an isolation requirement, shown in Figure 5.6. This longitudinal phase noise requirement led to triple suspensions, known as HRTS, being chosen to steer the LO and signal beams onto the combining beam splitter.

Chapter 6

Design of the LIGO A+ HAM6

Layout

The LIGO A+ HAM6 vacuum chamber will contain the readout photodiodes, the output mode cleaners (OMC)¹ and their associated steering optics. See Figure 2.4 for HAM6’s location with respect to the rest of the detector. Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector (BHD). The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout. As losses in the interferometer’s readout will degrade its sensitivity, it is vital that the active wavefront control (AWC) is designed in such a way as to minimise this.

In Section 6.1, the optical properties of the proposed HAM6 layout are examined. Section 6.2 is primarily a discussion of the design challenges associated AWC in HAM6. Section 6.3 is an analysis of how mode mismatches between the interferom-

¹The OMC assembly includes the readout photodiodes.

eter and the OMC can be corrected by the AWC. Section 6.4 is the conclusion for this chapter. This work was undertaken by a team made up of myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

6.1 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing (January 2021) is shown in Figure 6.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 6.2 . The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and the preferred design for the Balanced Homodyne Single Suspension (BHSS). The optical properties of this layout are summarised in Table 6.1. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as OMx1 and OMx2 as it is not important to make a distinction between the A and B paths when considering mode-matching.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 6.3. Significant points along the optical axis are also shown in this figure. The size of the beam at the final mode-matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirror's surface. The beam is always less than 1.2 mm within HAM6, an important fact to remember when considering how close structures can be to the beam path ².

²The minimise scattering, the clear aperture size for the beam is defined to be 10 times larger than the Gaussian beam width.

As astigmatism degrades the mode-matching, it can contribute to the loss in the readout of the detector. Non-zero angles of incidence (AOI) of non-astigmatic beams onto curved optics cause the reflected beam to become astigmatic (for a comprehensive description of astigmatism see [95]). For this layout, the astigmatic loss was calculated to be $< 0.05\%$; see Appendix E for detail in this type of calculation. The astigmatic loss is plotted as a function of angle in Figure 6.2; as the beam will have low curvature, the loss becomes significant when the AOI is above 10° . The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct; however, as some of the losses in the readout train of the interferometer are expected to be at the many percent level, losses of $< 0.05\%$ are considered unimportant.

The focal lengths of the OFI lens, OMx1 and OMx2 were optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible as this gives orthogonal control over the mode-matching between the arm mode and the OMC. This is shown in Figure 6.4. With the layout shown in Figure 6.1, it is possible to achieve 42° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, this layout offers many practical benefits, mainly in relation to the BHSS, over the other considered designs that outweigh an additional 3° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control as there is $> 80^\circ$ between OMxS and OMx2.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.066	-15.068	-
OMx1	-0.350	-2.855	5.709 (concave)
OMx2	-0.847	-1.180	2.360 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	500
OM0 to BHDBS2	946
BHDBS2 to OMxS	Path A: 225, Path B: 315
OMxS to OMx1	Path A: 1240, Path B: 1150
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \mu\text{m}$
OMx1 to OMx2 Gouy phase	42°
OMxS to OMx2 Gouy phase	Path A: 85° , Path B: 81°
OMx1 and OMx2 AOI	A: 9.7° , B: 4.9°

Table 6.1: Mechanical and optical properties of the LIGO A+ HAM6 layout.

In conclusion, the proposed HAM6 layout, shown in Figure 6.1, fulfils the criteria set in Section 6.2. Near optimal mode-matching and alignment control can be achieved with this layout, and a summary of the properties of the layout can be found in Table 6.1. The next stage of development for this layout involves the design and prototyping of parts and the consideration of fine details such as cable routing.

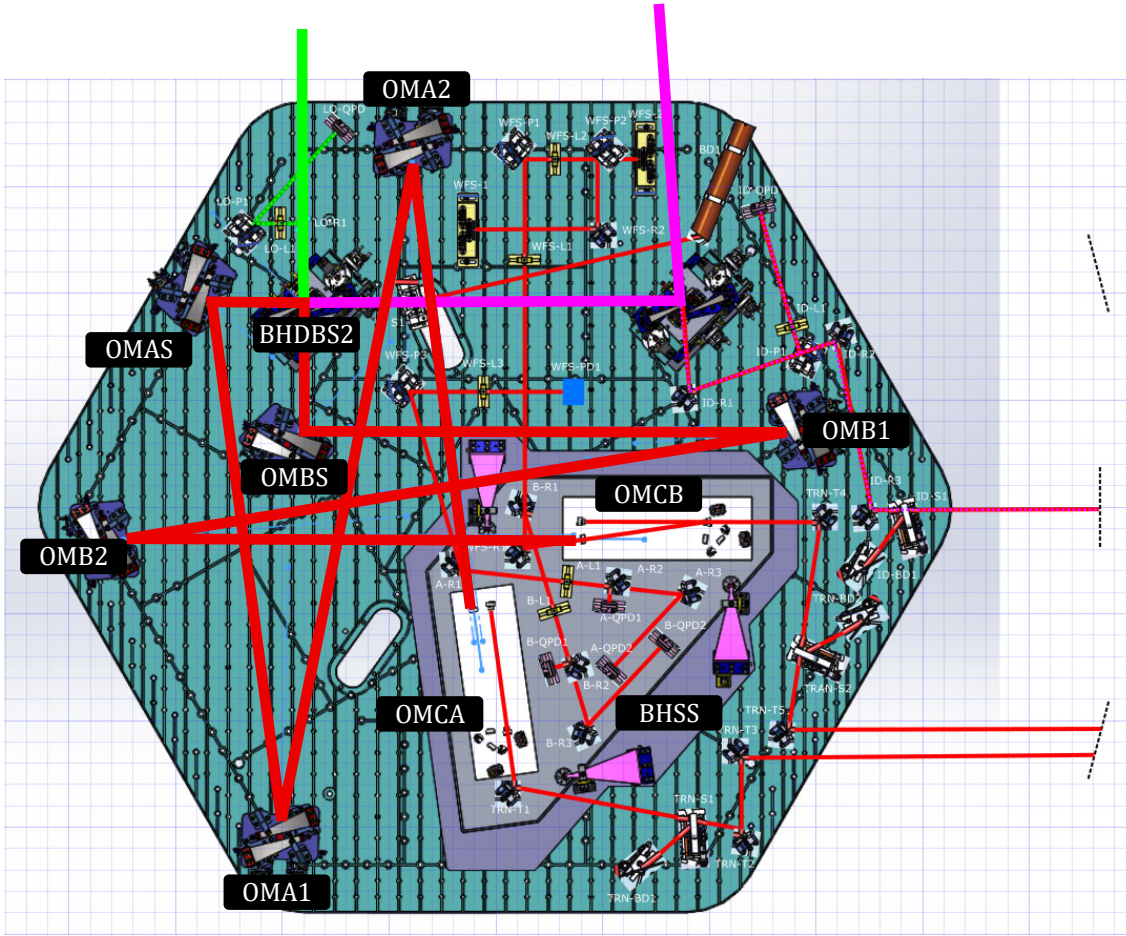


Figure 6.1: Place holder graphic. Caption to the effect of... The LO beam enters HAM6 directly above BHDBS2 (north side), and it is highlighted in green. A pick-off will be used to monitor the alignment of the LO; the pick-off beam is indicated with a dashed green line. The signal beam enters HAM6 directly above OM0 (north side), and it is highlighted in pink. The signal beam is steered by OM0 onto BHDBS2 where it is combined with the local oscillator. The shutter is located between OM0 and BHDBS2. A small amount of the signal beam is transmitted by OM0 (dashed pink); this beam is used align ???. The two combined beams are shown in red. These are incident upon a mirror used for steering (OMxS) and then upon two mirrors curved mirrors (OMx1 and OMx2). (Caption continued on next page).

Figure 6.1: (Figure on previous page). The BHSS (the large triangular component) takes up most of the bottom left of the HAM6 chamber. The position of the OMCs is shown. On the BHSS, there are several small optics used to steer beams that are required to control ??; the wavefront sensors (yellow components) for this are found at the top of the HAM6 table. Several beams are directed of the table for imaging with CCD cameras.

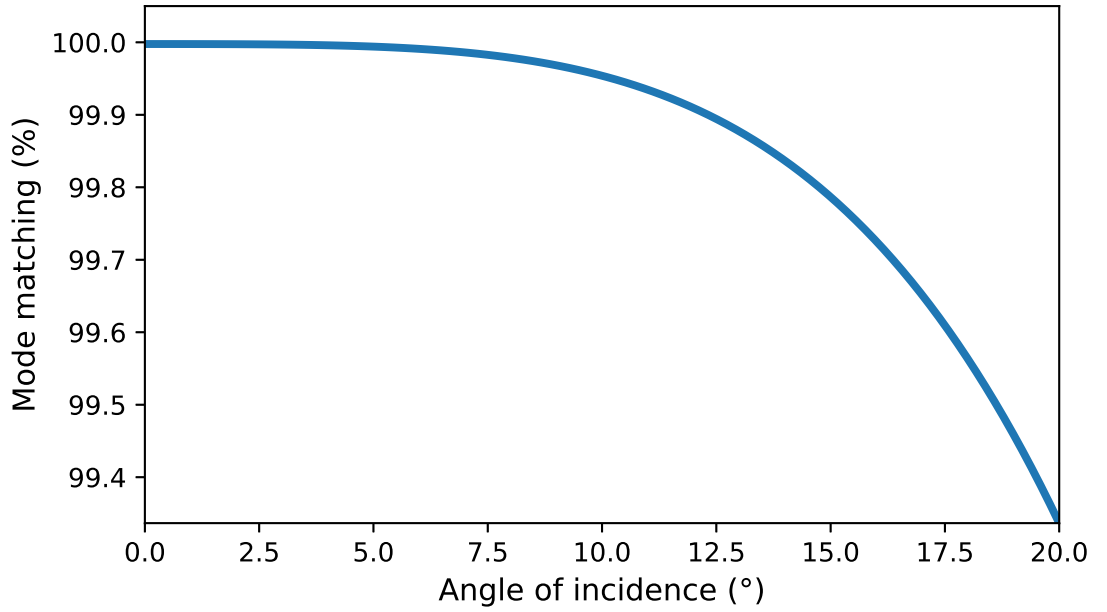


Figure 6.2: Astigmatism is a source of loss. When designing the layout shown in Figure 6.1, the AOI for the curved mirrors was altered to explore how sensitive it was to astigmatism. Using Equation (6.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 6.2.

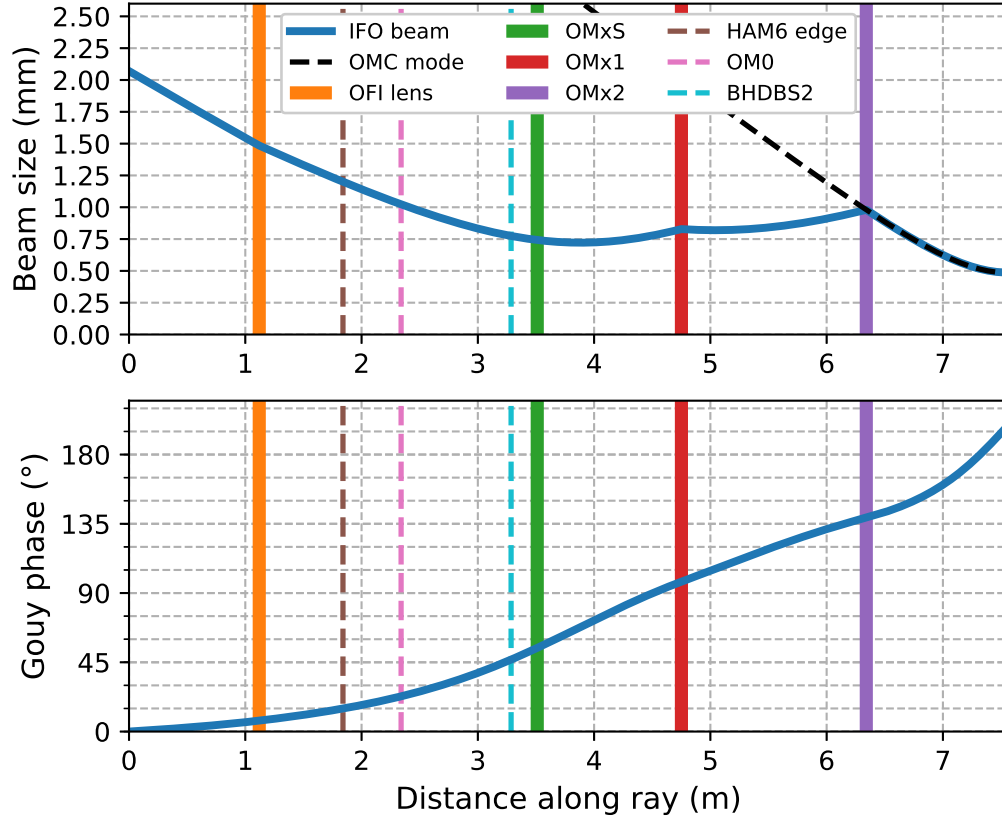


Figure 6.3: The top panel shows the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful, reference points are marked by dashed vertical lines. The bottom panel shows the Gouy phase shift accumulated by the beam as it propagates along the optical axis.

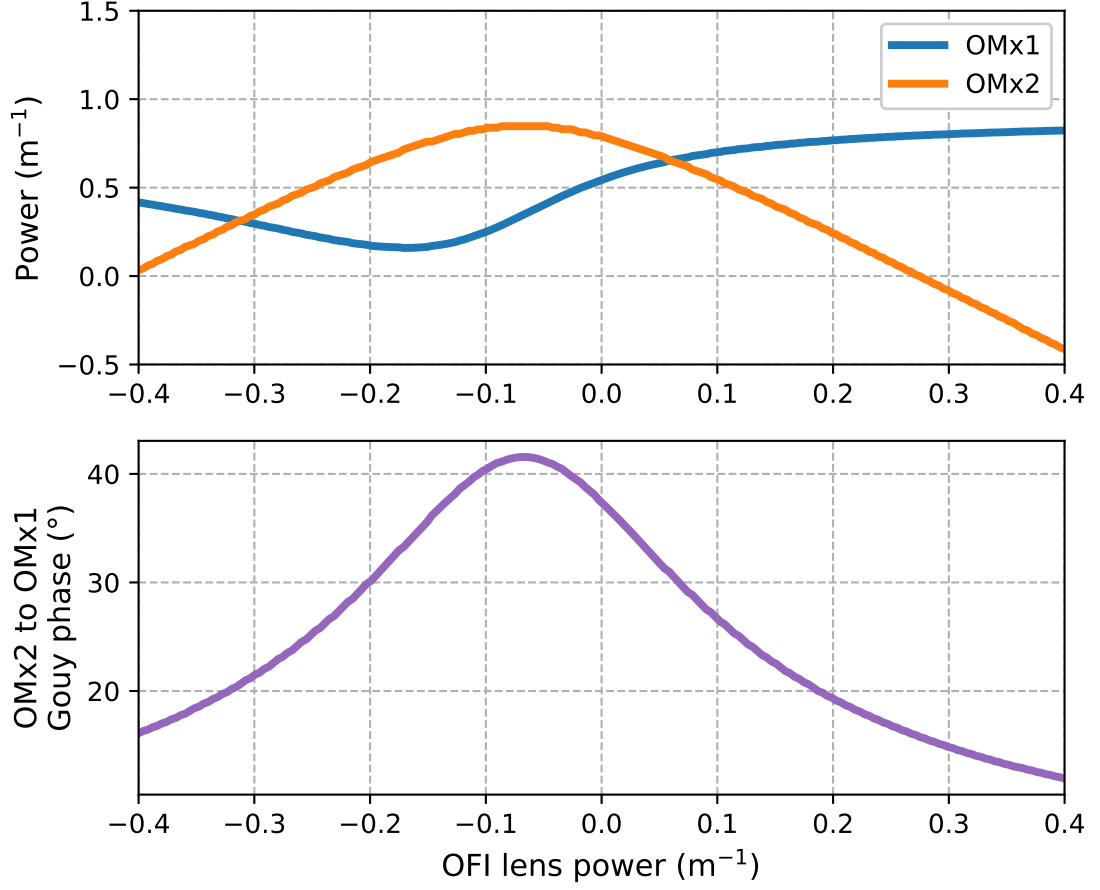


Figure 6.4: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power that is required for 100% mode-matching. Bottom panel: The best choice for of OFI lens that corresponds to obtaining orthogonal control of the beam parameter in HAM6 is where the Gouy phase separation between the pair of OMx1 and OMx2 is closest to 45° . The most Gouy phase separation between OMx1 and OMx2 achievable by this layout is 42° .

6.2 The Requirements for the Optical Layout Within HAM6

The BHD optical layout in HAM6 is required to provide a readout of the gravitational wave signal. To meet the design sensitivity of A+, HAM6 must house a layout that is robust to mode mismatch as there is significant uncertainty in the beam parameter of the arm cavity mode at the SRM. Having control over the two degrees of freedom associated with mode-matching – the size and location of the beam’s waist – between the OMC and the beam exiting the SRM was challenging as this requires a lot of space relative to the size of the optical table. In addition to this, the HAM6 layout needs to leave space for the other sensors and beams used for controlling the interferometer. All proposed designs used most of the table space. This section focusses on the design considerations relating to mode-matching; other aspects of the functionality required from HAM6 is summarised in [96].

To reach the desired level of squeezing for A+, the goal for mode-matching between the arm mode and the OMC mode is $> 98\%$; the loss contribution of the components in the readout train of the interferometer are summarised in Table 6.2 [70]. Active mirrors, called Suspended Active Matching Stages (SAMS), will be used to correct for mode-mismatch; they will provide at least ± 50 mD of optical power tuning range [97].

The AWC in HAM6 has two functions: it will be used to correct for thermal effects in other parts of the interferometer [96] such that the mode-matching between the arm mode and the OMC is maximised, and it will be used to correct for RoC errors

in the interferometer after the main beamsplitter.

Thermal changes in the arm cavity mode may be corrected for by the thermal compensation system (TCS) [98]. However, in practice operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, at LLO it has been found that TCS can be used to decrease laser intensity noise coupling to DARM [99]. As each interferometer subsystem may not be used for its intended purpose, having redundancy in the design of the interferometer is desirable if unexpected noise couplings are discovered. As TCS is used to decrease a noise coupling, it may not be able to correct thermally-driven errors in the arm mode. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

Two curved mirrors (OMx1 and OMx2) are required to control the two degrees of freedom associated with mode-matching. To gain flexibility in the design, a lens will be used to provide further Gouy phase separation between the two active mode-matching mirrors in each path after BHDBS2. There is space for this lens on the OFI assembly; this is vibrationally isolated and is a useful distance from the mode-matching optics in HAM6. An active lens was considered here, but concerns of OFI assembly heating [100] and the unavailability of suitable flint glass meant that this idea was discarded.

To provide orthogonal mode-matching control, the two SAMS need to be separated by 45° of Gouy phase. Due to the design of the OMC, the Rayleigh range of the beams in HAM6 are of the order of one meter. As most of the Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will

have to travel a significant distance across the table before it acquires sufficient Gouy phase. This Gouy phase separation requirement gave us a method of generating candidate designs as this requirement can be interpreted as needing the mode-matching mirrors to be as far apart as possible from each other on the HAM table. Losses due to astigmatism prevents one from using a large AOI to increase the separation between optics in HAM6.

Orthogonal alignment control is obtained by having two mirrors separated by 90° of Gouy phase. For all designs, the two mirrors for alignment were immediately after BHDBS2 and immediately before the OMC as this gave the best Gouy phase separation between them. It was found that placing the first mirror close to BHDBS2 often provided good layouts. References [101, 102] describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled.

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer [96, 103]; as described in [103], backscatter can degrade the sensitivity of the interferometer. A single stage suspended platform (BHSS) is required to prevent backscatter noise from reducing the interferometer's sensitivity. Both OMCs will need to be on this platform and so all designs need to have the OMCs fairly close to each other. A large BHSS is mechanically infeasible as it will reduce space for the other components on the HAM6 table.

6.2.1 Fixed properties of the HAM6 Optical Layout

There are a few properties of the HAM6 layout that will not be changed. These include: the entry point of beams into the HAM6 chamber, the size of the nominal

mode entering HAM6 and the size of the OMC waist. Table extensions can be made in places where they would not impede access, however the size of the HAM optical table does restrict the design. It was decided that the two beam paths after BHDBS2, the optic on which the LO and signal is combined, would be as similar as possible as both beams need to be matched to OMCs of the same design.

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [104], so the design of the HAM6 mode-matching optics is based on this quantity. As discussed in Chapter 4, the SRM beam parameter is expected to differ from its nominal value. It is described in Chapter 4 how geometry errors in the SRM cause the arm cavity and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure ??, was taken into consideration when choosing a layout for HAM6 since different candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The OMC design developed for aLIGO [103], with trivial modifications, will be used to remove junk light from the signal beam and so there is no choice in target waist size. The nominal OMC waist size is $485 \mu\text{m}$. The beam enters the OMC through a flat mirror, and a waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode-matching optics therefore have to match the nominal SRM and OMC modes.

HAM6 loss Source	Target for O5
OMC throughput loss	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 6.2: Target for losses in HAM6. Reproduction of [70]

6.3 Visualisations of the Robustness of the A+ HAM6

Layout to Mode Mismatch

As described in Section 6.2 and Chapter 4, it is expected that there will be some mismatch between the arm mode and the OMC. To visualise the effectiveness of the AWC provided by the layout’s active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode-matched from by the AWC and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout.

The proposed layout in Section 6.1 can mode-match the majority of beam parameters one would expect based on uncertainties in the RoC of the optics within the SRC. Despite this good coverage, measurements from LLO show that the beam parameter may fall outside the region of WS which the layout can compensate for as the LLO SRC differs significantly from the nominal design. Alternative optical powers for the OFI lens, OMx1 and OMx2 are given in Section 6.3.5; this change would sacrifice orthogonality of the mode-matching and alignment optics in favour of robustness to RoC errors in the SRC.

6.3.1 A Description of Mode-Matching in Terms of HG Modes

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode-mismatch. However, mode-matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are. Mode-matching between two modes is usually, to first approximation, thought of as a difference in their waist size and waist location.

Small mode-mismatches can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $\text{HG}_{20} + \text{HG}_{02}$ modes. If a cavity is designed such that the fundamental mode will be resonant while the $\text{HG}_{20} + \text{HG}_{02}$ modes will be reflected, the effect of mode-mismatch between the cavity and a signal carrying beam is to decrease the signal light that is transmitted by the cavity since some of the signal carrying light will be scattered into the reflected $\text{HG}_{20} + \text{HG}_{02}$ modes. Sensing the $\text{HG}_{20} + \text{HG}_{02}$ modes therefore gives a measure of the mode-matching between two modes.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$ modes, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ modes which is 90° out of phase with the fundamental mode [105]. As the $\text{HG}_{20} + \text{HG}_{02}$ modes acquire 2ϕ compared to the fundamental mode, where ϕ is the Gouy phase acquired by the fundamental mode, spacing beam curvature actuators at 45° will give an orthogonal basis for control over mode-matching. Orthogonal control results in the beam size and beam waist,

or two combinations of them that are orthogonal, can be independently controlled.

6.3.2 Calculating Mode-Matching Using The Gaussian Beam Parameter

Two perfectly mode-matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (6.1)$$

can be used to determine the amount of mode-matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*||q_{y2} - q_{y1}^*|}, \quad (6.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. This is derived in Appendix E. The solution to Equation (6.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode-matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [106]. The mode-matching can be calculated at any point along the optical axis of the layout as mode-matching must be preserved throughout it; however, it makes sense to calculate the mode-matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

6.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S [107]. A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirror's surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (6.3)$$

From Equation (6.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise [108, p.680]. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode-matching and the effect of actuators on the beam size

(rather than $\lambda/\pi w^2$) and defocus, the WS representation was chosen. One could express the beam parameter in terms of natural units by dividing the beam size by w_0 and the defocus by $1/z_R$.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 6.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode-matching requirement between the interferometer and the OMC for LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode-matching between two arbitrary modes can not be deduced from these contours [107]. For example, two modes may be mismatched by the same amount from the target in orthogonal directions in WS space. They cannot be 100% mode-matched as these two modes are different from each other.

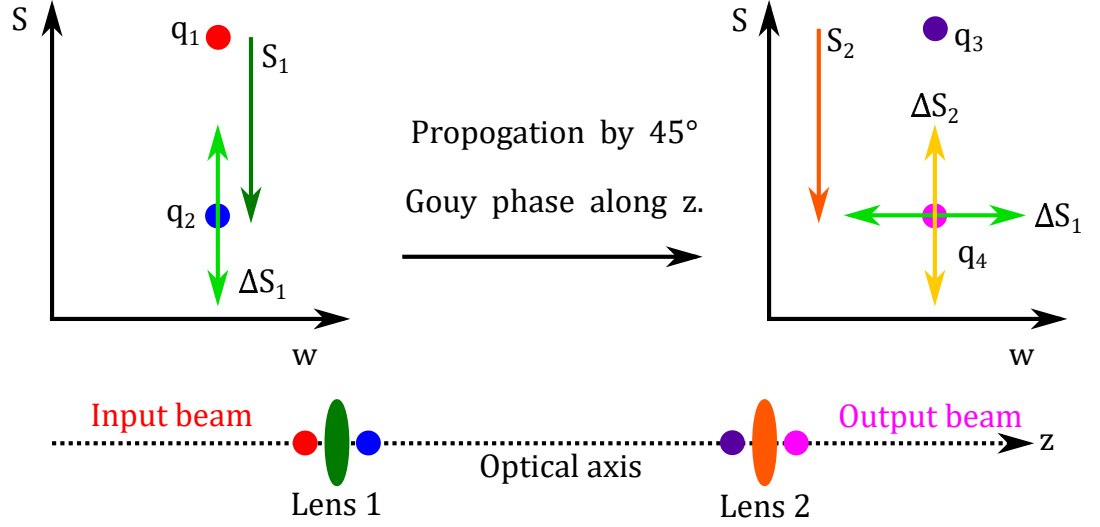


Figure 6.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axes shows the WS plane at Lens 1. The input beam, q_1 , is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of Lens 1. The right hand axes shows the WS plane at Lens 2. The input mode at Lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of Lens 2 is indicated by the orange and yellow lines. As Lens 1 and Lens 2 are separated by 45° of Gouy phase, the actuation provided by Lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by Lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and the location of the modes.

6.3.4 Visualising how Active Mode-Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode-matching space can be achieved was computed. To do this, first the ring of modes in WS space that is 98% mode-matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode-matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it whereas for the lower region the actuation should be $-\Delta S$. This results in a region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode-matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the region. This process is shown in Figure 6.6.

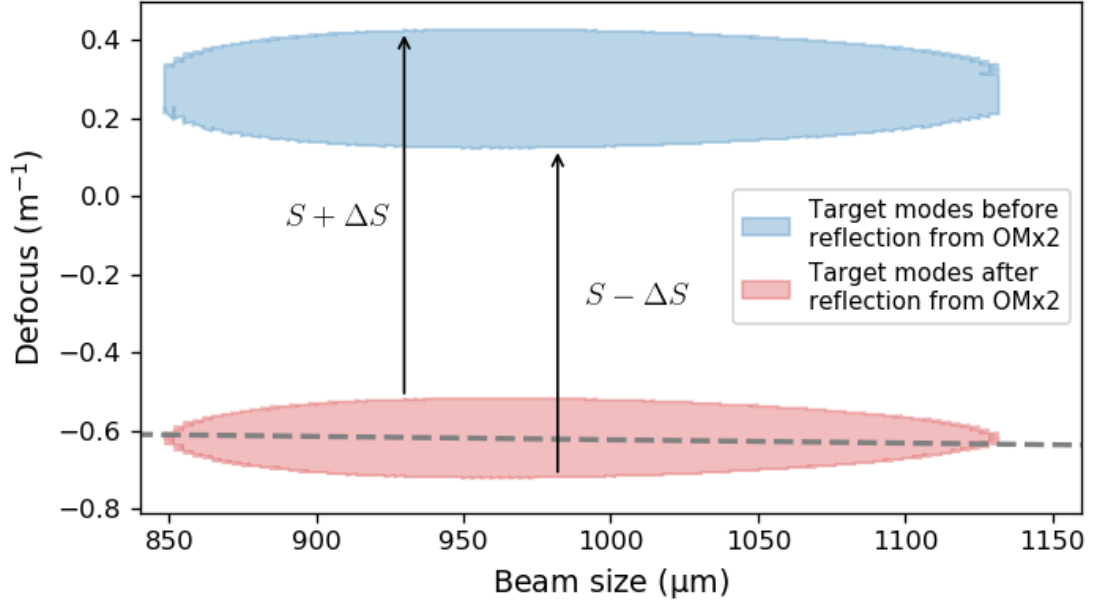


Figure 6.6: The WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode-matched to the OMC as there are no further active mode-matching elements in the layout of HAM6. Modes with at least 98% mode-matching are shown by the red region. The blue region represents modes which can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

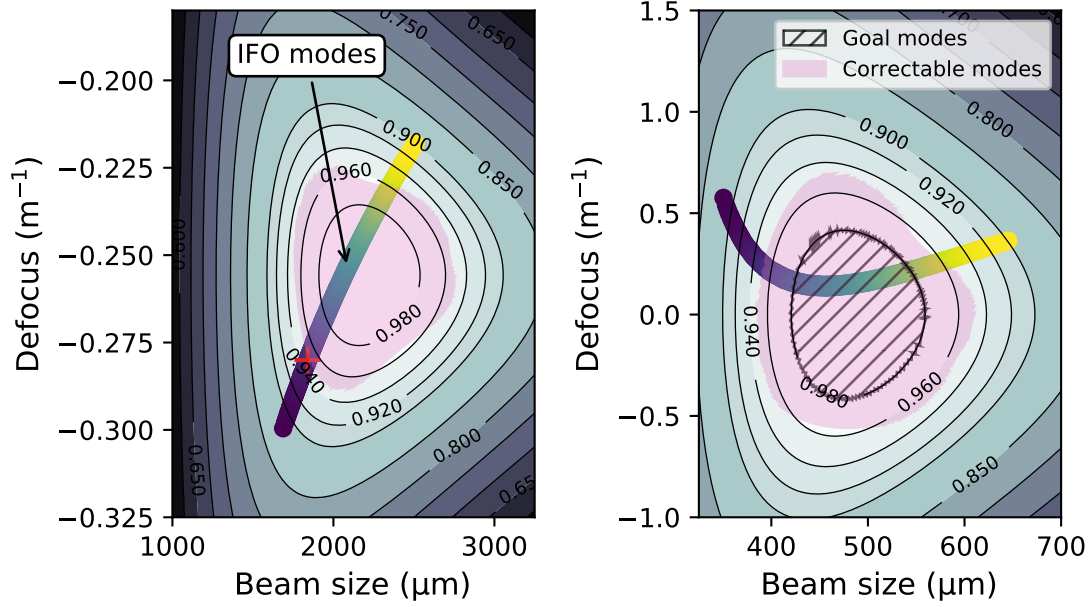


Figure 6.7: The region of WS space covered by the AWC in HAM6 if OMx1 and OMx2 for the layout discussed in Section 6.1. Left panel: the set of modes that is expected to be present at the SRM, as described in Chapter 4, is shown by the purple-blue-green-yellow curve. The pink region shows which modes that can be corrected for so that 98% mode-matching is achieved. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. The red cross indicates the mode size that can be inferred from measurements of the SRC's Gouy phase at LLO. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

The modes from the SRM that can be matched to the target are shown in Figure 6.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter 4), the interferometer modes were compared to the region of WS space containing modes which can be corrected such that they meet the $> 98\%$ mode-matching requirement. It was found that majority of likely modes will be covered by the layout as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre; however, measurements of the SRC's Gouy phase at LLO can be used to infer the beam parameter at the SRM, and this mode would fall outside of the region of WS space which can be compensated for by the AWC. Its worth noting that the modes at the yellow end of the continuum shown in Figure 6.7 may not be physical as they correspond to an unstable SRC.

The shape of the pink region in Figure 6.7 is nominal; when the HAM6 layout is constructed the pink region's shape will be altered by geometric errors in OMx1 and OMx2. Nevertheless, Figure 6.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode-matching errors generated by errors in the SRC.

6.3.5 Optimising WS Space Coverage in Favour of Expected Mode Errors

While 45° of Gouy phase separation between the mode-matching optics in HAM6 is a desirable quality for the layout to have, with an actuation range ± 50 mD, the layout can be optimised to correct for mode-mismatches between the OMC and the continuum of expected beams exiting the interferometer. The majority of the

uncertainty in the interferometer's beam parameter comes from the uncertainty in SR3's RoC (see Figures ?? and ??). Ideally, one would correct for this by actuating on the RoC of SR3, however the RoC of SR3 cannot be strongly actuated on. If a mirror was placed 180° away from SR3, it could be used to compensate for the uncertainty in the RoC of SR3 i.e. one could configure the mode-matching optics within HAM6 such that they are biased for correcting the errors in the interferometer beam parameter caused by the error in SR3.

By reducing the orthogonality of the AWC actuators, more range can be obtained for one of the beam's property e.g. the size of a beam's waist. A Gouy phase separation ($\sim 30^\circ$) offers a compromise between orthogonality versus actuation range for errors in one of the beam parameters dimensions. Figure 6.8 shows how the region of WS space covered by the HAM6 AWC can be biased in favour of the expected set of modes that may emerge from the SRM. The physical positions of the optics are the same as before (see Table 6.1), however the optical powers of the OFI lens, OMx1 and OMx2 was altered (-0.2, 1.67, 0.631 respectively) to change the Gouy phase separation between OMx1 and OMx2 to 30° . However, the Gouy phase for alignment control would be significantly reduced; this is shown in Figure 6.9.

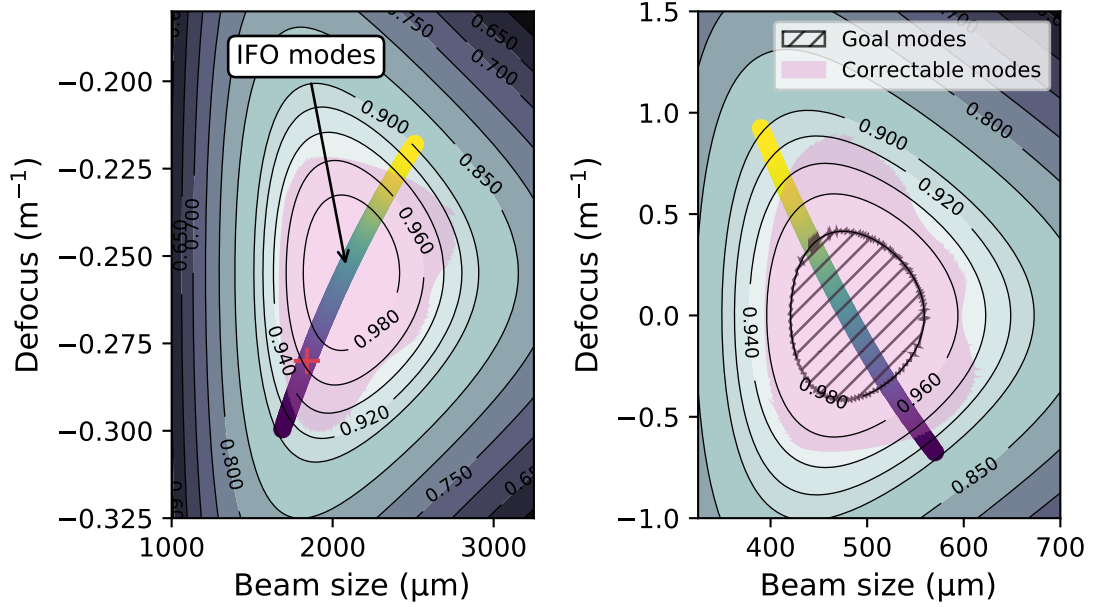


Figure 6.8: At the expense of orthogonality, the region of WS space covered by the AWC in HAM6 can be optimised by changing the Gouy phase separation between OMx1 and OMx2. Left panel: the set of modes that could be emerge from at the SRM, as described in Chapter 4, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel; however, they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region (hatched area).

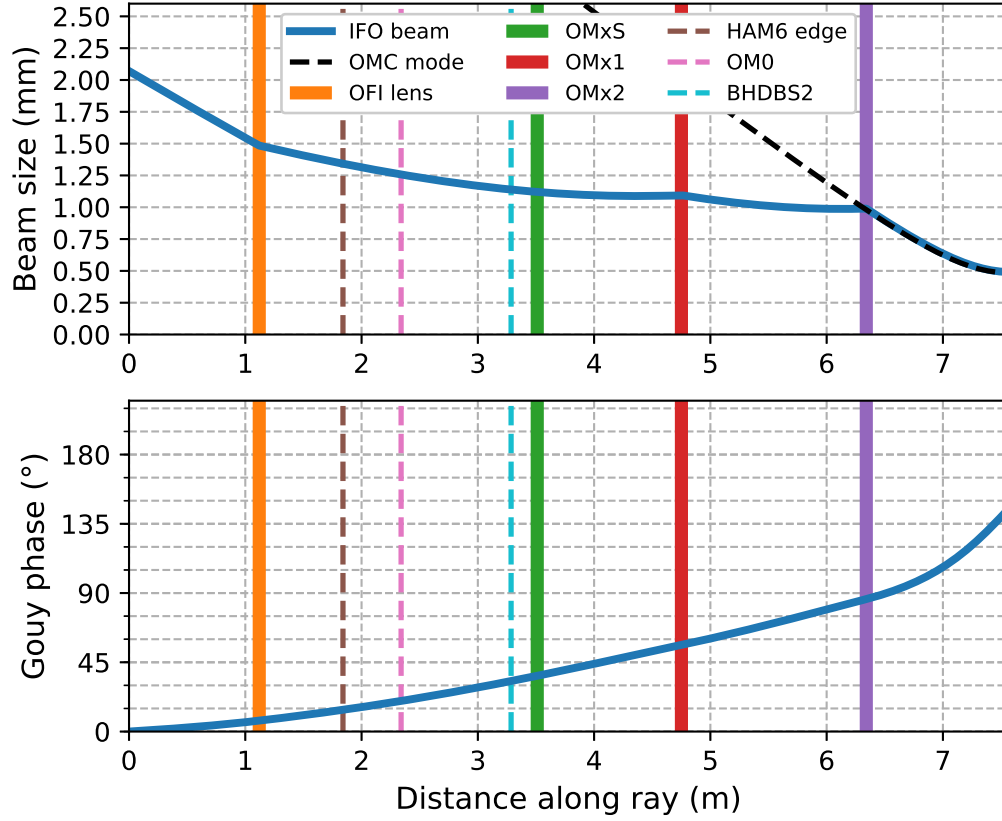


Figure 6.9: Top panel: the beam profile for the alternative layout of HAM6 suggested in Section 6.3.5. Bottom panel: The Gouy phase accumulated along the optical axis for the alternative layout. While alignment Gouy phase is degraded, the mode-matching Gouy phase was changed to bias the optical layout for correcting errors in the RoC of SR3.

6.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements was presented in Figure 6.1. The optical properties of this layout were investigated. By selection of the right combination of mode-matching optics, near orthogonal control of the signal-LO mode-matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode-matching requirement space was investigated. WS diagrams were used to represent how robust the layout is to initial mode-mismatch. Figure 6.7 shows the amount of WS space that the AWC could correct for. It would be desirable to shape this region to improve the chances of being able to mode-match the interferometer to the OMC. At the expense of AWC and alignment orthogonality, the mode-matching optics in HAM6 could be changed from their nominal values to improve the likelihood that the AWC can correct for mode mismatches as described in Section 6.3.5.

Chapter 7

Experimental MZ

Chapter 8

Characterisation of InGaAs

Photodiodes for detecting 2 μ m Light

As discussed in Section 2.8 and 2.10, a new photodiode technology will be required in the readout train of the third generation of gravitational wave detector. Quantum efficiency, power handling and dark noise requirements are three important properties for a photodiode that would be used as for readout of gravitational wave detectors; the requirements for these properties is explored in Section 8.1. Extended InGaAs photodiodes are sensitive to 2 μ m light; however, for the reasons explained in Section 8.2, fabrication of an extended InGaAs photodiode that meets the requirements set out in Section 8.1 is challenging. The behaviour of off-the-shelf photodiodes was measured under different reverse biases to explore this. Alternative photodiode materials that could be used include HgCdTe and InSb. HgCdTe photodiodes that are optimised for 2 μ m light with quantum efficiencies $> 99\%$ are in the design stages of being made, however there are still manufacturing challenges to overcome [125]. InSb detectors have high quantum efficiency [126], however this material would only

be suitable for wavelengths above $2.6\,\mu\text{m}$ due to its quantum efficiency being >1 below this wavelength.

8.1 Requirements for the Readout Photodiodes in a Gravitational Wave detector

Quantum efficiency, power handling and dark noise requirements are three important properties for a photodiode that would be used as for readout of gravitational wave detectors. As discussed in Section 2.8 and 2.10, a new photodiode technology will be required in the readout train of the third generation of gravitational wave detector.

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency $> 99\%$ [70]. Loss within the interferometer decreases its SNR as it diminishes the level to which the signal light is squeezed (see Equation 2.17). High quantum efficiency is required to detect the baseline level of $\sim 10\,\text{dB}$ of squeezed light that future gravitational wave detectors will use [77,109,115]. Usually, if a bigger signal is required, the gain can be increased by using a larger resistor in the feedback path of the transimpedance amplifier or by using an avalanche photodiode, however neither of these methods would work in a gravitational wave detector as both would boost the noise and the signal in proportion.

Current gravitational wave detectors operate in the $1\text{-}10\,\text{mW}$ range at the output [49] to ensure that the junk light from other modes is a small portion of the total signal. As $q\lambda_{\text{EM}}/hc$ is of the order unity, the photocurrent will be assumed to be $\sim 10\,\text{mA}$ and so any new photodiode should be able to produce $\sim 10\,\text{mA}$ of

photocurrent without saturating. If heterodyne readout was to be used, then the DC light power could be reduced.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that the interferometers are not limited in their detection bands by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [115, 142], so the $1/f$ noise must be below the shot noise of the photocurrent above this frequency. For a 10 mA photocurrent, the shot noise is $6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$. Below 10 Hz, other rapidly rising sources of noise will limit the sensitivity of the Einstein Telescope [75, 115] and Cosmic Explorer [109]

The $1/f$ noise could be avoided by using heterodyne detection, however a conventional heterodyne detection scheme would increase the shot noise level of the signal [83, 143, 144]. Heterodyne detection can decrease the SNR of the interferometer if the mixing of the modulation and demodulation waveforms was not time independent [83]. Although an optimal choice of modulation-demodulation waveform can recover the SNR by increasing the signal, the benefits of using squeezed light would be lost because the shot noise would be enhanced compared to homodyne readout by vacuum noise at frequencies that does not contain signal entering into the measurement. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [145].

8.2 What is an Extended InGaAs Photodiode

In this section, a qualitative explanation of what an extended InGaAs photodiode is and how photodiodes work will be given. More details on the semiconductor physics of diodes and photodiodes can be found in many textbooks e.g [127, 128]. The relationship between defects in InGaAs photodiode and properties such as dark current, dark noise and quantum efficiency of photodiodes will be discussed.

The basic physics of a PIN photodiode can be qualitatively understood by considering a simpler pn junction. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the p-doped region, holes (the majority charge carriers) are free to move and there are negatively charged ions (acceptors) that are fixed in position. In the n-doped region, electrons (the majority charge carriers) are free to move and there are positively charged ions (donors) fixed in position. In equilibrium, the number of holes is equal to the number of electrons. The free electrons/holes will diffuse across the interface between the p-doped and n-doped region. This causes the p-doped and n-doped regions to become charged. This results in a region of the pn junction where there are no free electrons and holes. The region of the device containing no free carriers is called the depletion region.

When a photon is absorbed by a pn junction, a free electron-hole will be generated. If this photon is absorbed within the depletion region of the photodiode, the electron and hole will be separated due to the electric field and will generate a current in the circuit connected to the photodiode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will recombine and will not contribute to the current in the circuit. The quantum efficiency of a photodiode is the proportion

of photons that are converted into photocurrent. The relationship between quantum efficiency, η , light power, P_{EM} , and photocurrent, I_p is

$$I_p = \eta \frac{q\lambda_{\text{EM}}}{hc} P_{\text{EM}}, \quad (8.1)$$

where λ_{EM} is the wavelength of the light, h is Planck's constant and c is the speed of light.

To widen the depletion region, an external voltage can be used. When a *reverse bias* – a positive voltage across the cathode and anode – is applied to the photodiode, there is a larger electric field difference at the interface between the p-doped and n-doped region; thus the depletion region is widened i.e. applying a reverse bias to a photodiode increases the volume of the photodiode in which absorbed photons will generate photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p-doped and n-doped region. An intrinsic semiconductor has no doping and the number of electrons equals the number of holes within it. As the intrinsic region physically separates p-doped and n-doped regions, a larger amount of the diode will be depleted of electrons and holes due to the electric field within the diode. Additionally, the capacitance of the diode will be decreased.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$,

so that the InGaAs can be lattice matched to the InP substrate¹ on which the InGaAs is grown. With this amount of indium, the InGaAs photodiode will suffer from few lattice mismatch defects and so the photodiodes created with it will be of high quality. Normally they are sensitive up to wavelengths around 1.6 μm .

Longer wavelength sensitivity can be achieved by increasing the indium content of the photodiode [130]. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 3 μm . One of the main issues faced when creating an extended InGaAs photodiode is the strain induced by lattice mismatch between the InGaAs and the InP substrate [131] as this introduces defects.

Defects caused by the strain between the InGaAs and InP layers of the photodiode generate excess noise, dark current and reduce its quantum efficiency [132]. The photodiode can be made thinner to reduce this strain, although this reduces the quantum efficiency. The critical thickness, that is the thickness above which strain degrades the crystal, for extended InGaAs is of the order 0.1 nm [131] whereas the absorption length (of the order of 10^4cm^{-1} [133]) requires the photodiode to be several microns thick for it to have the high quantum efficiency that is required for the detection of gravitational waves (see Section 8.1). Additionally, because InGaAs is not a good conductor of heat, a thicker amount of InGaAs can lead to a lower power threshold before the photodiode is thermally destroyed [134].

These defects manifest as abrupt changes in the crystal structure known as misfit dislocations (e.g. [135] [136]). Defects produce trap states and these give rise to a generation-recombination dark current [137]; thus defects are the source of 1/f

¹Ge can also be used

noise [138]. To obtain high quantum efficiency the defect level must be low [139]. Defects increase the doping level of the absorbing layer of the photodiode which causes the depletion region to become smaller, thus causing a decrease in quantum efficiency. For indium doping levels other than 53%, buffer layers of InGaAs between the absorbing layer and the substrate can be used to mitigate the lattice mismatch [137].

InGaAs that is optimised for 2 μm has a refractive index, n , of approximately 3.6 [133]. Assuming a single quarter-wave layer was to be applied, the film would need a refractive index that was $n_f = \sqrt{3.6}$ and be 600 nm thick. The idea of using this anti reflective coating as a passivation layer is presented in [140]. Some manufacturers create a range of photodiodes with different spectral responses by changing the anti-reflective coating rather than changing the photodiode's indium content [110].

The dark noise and dark current of a photodiode has several contributors. In low bias regimes, the dark current is driven by the diffusion of minority carriers [139,141]. However, the thermally driven generation-recombination of electron/hole pairs drives the dark noise at low biases [138]. Therefore, the dark current does not predict the dark noise, however as both of them increase with higher levels of defects, a photodiode with a high dark current will have a high dark noise. When the bias is large enough, electrons can tunnel across the junction leading to a great increase in dark current.

The noise of the current made by the photodiode can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R_{\text{PD}}} + \left(\frac{i_c}{f^\beta}\right)^2}. \quad (8.2)$$

The first term under the square-root, $2qi$, is the shot noise associated with the DC current i . The second term, $\frac{4k_B T}{R_{\text{PD}}}$, is the Johnson-Nyquist noise due to the shunt resistance of the photodiode, R_{PD} ; this term can become significant when the shunt resistance is low, as it is in some extended InGaAs photodiodes, and microwatt levels of light is being detected. The final term is Hooge's law of noise in semiconductors [138]; i_c is the characteristic noise level and β is usually 0.5 ± 0.1 .

8.3 Selected Off-the-shelf Extended InGaAs Photodiodes

The dark current and dark noise was characterised in a selection of commercially available photodiodes. The quantum efficiency as a function of bias was measured for one of the photodiodes, the FD10D, and the saturation limit of the Laser Components photodiodes was investigated. The photodiodes that were tested and their properties according to their datasheets² are listed in Table 8.1. According to each

²Datasheets often quote the noise equivalent power (NEP) of a photodiode. The NEP is the noise spectral density of the dark current generated by the detector divided by the responsivity of the photodiode. For the Hamamatsu g12183 series photodiodes, the responsivity is $\sim 1 \text{ A/W}$ and the NEP is $\sim 10^{-12} \text{ W}/\sqrt{\text{Hz}}$, so the noise, as claimed by the datasheet, is $\sim 10^{-12} \text{ A}/\sqrt{\text{Hz}}$. This would seem sufficiently low to be shot noise limited down to $\sim 1 \mu\text{W}$. However, this number does not capture the behaviour of the photodiode at illumination levels of several milliwatts where a bias is required to ensure the photodiode does not saturate. As the noise of these photodiodes is not flat (see Appendix ??), NEP should be specified as a function of frequency. The noise and responsivity are both functions of reverse bias and so the NEP is a function of bias. Another quantity related to the NEP is the specific detectivity; this incorporates the area and bandwidth of the photodiode with the NEP.

Photodiode	Peak quantum efficiency (%)	Cut-off wavelength ^a (μm)	Diameter (mm)
FD10D	80 ^b	2.5	1
G12183-010k	77 ^b	2.5	1
G12183-005k	77 ^b	2.5	0.5
IG26x500S4i	86 ^c	2.5	0.5
IG24x500S4i	84 ^c	2.4	0.5
G12182-005k	84 ^d	2.1	0.5
G12182-010k	84 ^d	2.1	1.0

^a λ_{EM} for $\eta = 50\%$ ^b $\lambda_{\text{EM}} = 1.9\mu\text{m}$ ^c $\lambda_{\text{EM}} = 2.0\mu\text{m}$ ^d $\lambda_{\text{EM}} = 1.6\mu\text{m}$

Table 8.1: Datasheet information for the photodiodes that were tested. Note that none of the photodiodes has a quantum efficiency $> 99\%$, however measurements of dark noise and dark current and investigations into their optimal operating conditions is still of interest. This selection of photodiodes covers a range of manufacturers, sizes and cut-off wavelengths.

of the photodiodes datasheets, none of them have a quantum efficiency of $> 99\%$; these photodiodes have quantum efficiencies around 80%. However, the quantum efficiency of a photodiode can be improved by increasing the bias voltage (see Section 8.4).

Increasing the bias increases the dark current and dark noise of the photodiode and so for each photodiode a maximum reverse bias voltage was found. The benchmark used was the photodiode's dark noise at 10 Hz; the maximum useful bias was determined by the voltage where the dark noise was equal to the shot noise of a 10 mA current. The bias voltage also affects the saturation limit of the photodiode. If a photodiode was found to have a low saturation threshold (~ 1 mA) then it would not be fit for use in a gravitational wave detector.

8.4 The Effect of Reverse Bias on the Quantum Efficiency of an Extended InGaAs Photodiode

The relationship between the reverse bias and quantum efficiency of an extended InGaAs photodiode, the Thorlabs FD10D, was investigated. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: one where the reverse bias and the response of the photodiode were measured at DC and a lock-in style measurement where the quantity $\frac{\Delta\eta}{\Delta V_B}$ was measured. The quantum efficiency should be independent of frequency if the photodiode is operating in a time-independent, linear fashion and so the two styles of measurements should yield the same outcome. To prove that the increase in signal was a result of the photodiode's quantum efficiency increasing, and not some other mechanism e.g. an avalanche effect, the photodiode was biased such that the dark noise was sufficiently low (see Section 8.5 for details.) and the noise of the photocurrent was compared to the level expected for shot noise.

8.4.1 Method

To measure the photocurrent produced by the photodiode, a transimpedance amplifier which could produce shot noise limited signals for 10 mA photocurrents was used. This circuit is shown in Appendix ???. The circuit allowed for the reverse bias of the photodiode to be set externally via a low-passed bench-top power supply.

A 1550 nm wavelength laser (Pure Photonics PPCL300) was used to illuminate the FD10D. The light from the laser was split by a 50:50 beam splitter so that the laser could be amplitude stabilised. The in-loop photodiode was a high performance

InGaAs photodiode (C3061GH). The FD10D was the out-of-loop photodiode. The laser was amplitude stabilised by actuating on the laser photodiode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D and a lens was used to set the size of the beam.

The FD10D was exposed to a variety of beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the size and power of the beam. As the FD10D has a diameter of 1 mm, the laser beam should not have a width larger than $330\text{ }\mu\text{m}$ for there to be $< 1\%$ losses due to clipping. As gravitational wave detectors operate with 1-10 mW of power, the FD10D was tested with powers in this range. The methods in this section can only be used to measure relative increases in quantum efficiency; measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. It is straightforward to determine the change in quantum efficiency by measuring the change in photocurrent at DC. Alternatively, one can perform a lock-in style measurement. If the quantum efficiency depends on the reverse bias of the photodiode, modulating the reverse bias, V_B , at a frequency f will result in η modulating at f and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation is $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode and θ is the phase between the input and output modulation.

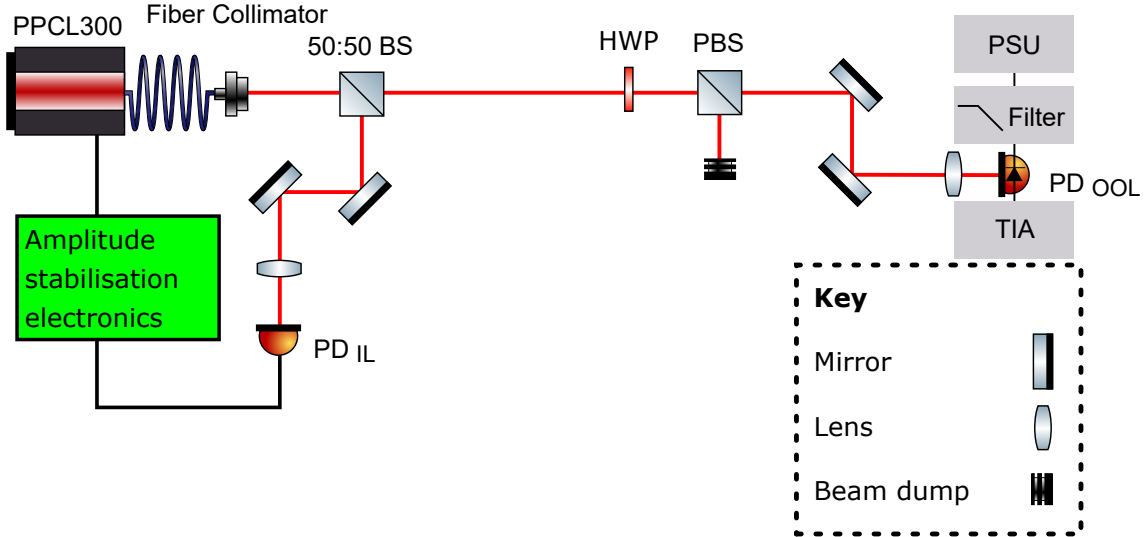


Figure 8.1: Optical layout used for measuring the effect that increasing the reverse bias of a photodiode has on its quantum efficiency and saturation limit. The out-of-loop photodiode, PD_{OOL} , was the photodiode under test. A 1550 nm laser was used to test the optical properties of the photodiode. A combination of half wave plate (HWP) and polarised beam splitter (PBS) allowed for the power of the beam to be set and a lens was used to set the size of the beam on the photodiode under test. Amplitude stabilised light was needed to ensure the increase in photocurrent seen when the bias increased was due to an increase in quantum efficiency. The photodiode circuit (TIA) (see Appendix ??) allowed for shot noise limited measurements of currents around 10 mA and for the bias to be selected.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see Figure 8.5), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. The expected level of noise, n , in units of $A_{\text{rms}}/\sqrt{\text{Hz}}$, is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (8.3)$$

where q is the charge of the electron, $I_{\text{OOL/IL}}$ is the current through the out-of-loop and in-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop photodiodes and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser} is the noise of the analyser.

The light incident onto the FD10D was amplitude modulated at 1111 Hz. This modulation was used to calibrate the measurements of noise as the relative intensity of this peak was known and could be checked against the signal measured by the high performing in-loop photodiode. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the saturation limit of the photodiode could be investigated.

8.4.2 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 8.2. For all beam intensities that the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the FD10D's datasheet. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced

a greater increase in η than beams with a higher intensity, however the quantum efficiency may not be uniform over the photodiode – the spatial dependence of η can be crater-like with η being greatest at the edges of the photodiode. The difference between the DC and AC response could only be described by a non-linear process. These effects are small and Figure 8.2 gives a clear incentive to increase the bias voltage to improve the quantum efficiency of the FD10D.

Figure 8.3 shows the photocurrent measured has the noise level predicted by Equation 8.3. This means that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. At 0.6 V reverse bias, for DC photocurrents ranging from 7.5 mA to 15.5 mA, the calibration peak and the noise level of the photocurrent were measured to ensure the photodiode was not saturating. Over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 8.4. The ratio changed in the opposite manner to how one would expect if the photodiode was saturating; however, The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak and the level of shot noise and so this change is not significant.

Taking into account the light reflected by the FD10D's window, at 0.6 V reverse bias, the quantum efficiency was calibrated using two other power meters. The internal quantum efficiency was measured to be 70%. There is a significant amount of light reflected by the FD10D's window; if the window is not removed, at 0.6 V reverse bias the quantum efficiency would be 60%.

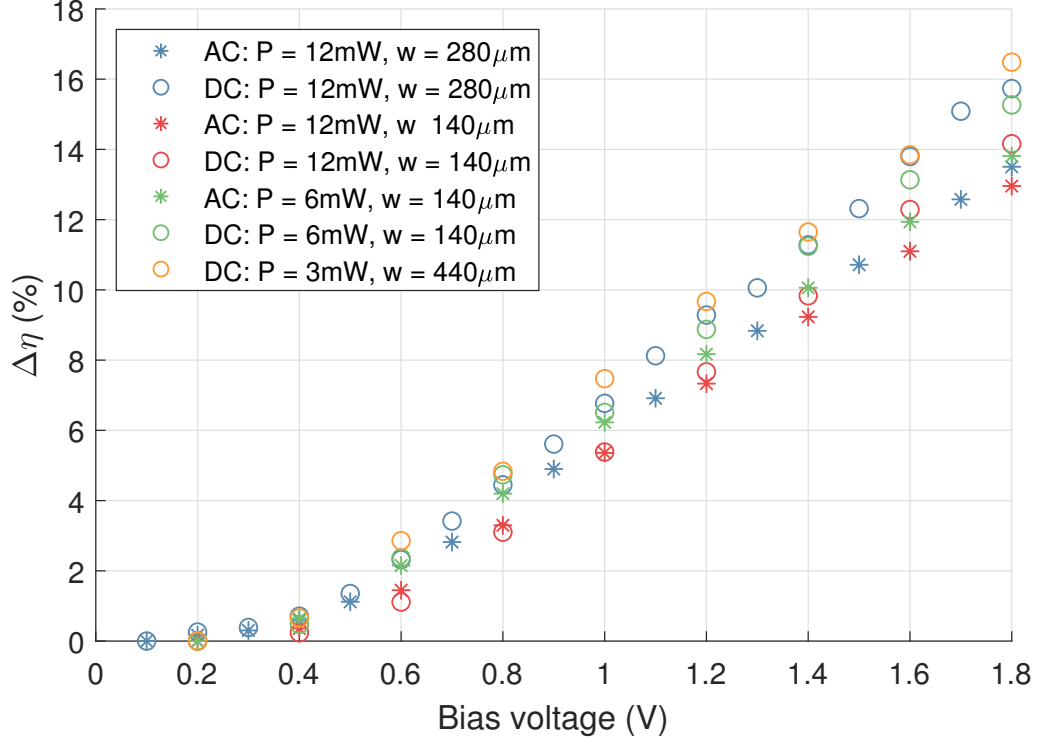


Figure 8.2: The FD10D's quantum efficiency depends on its reverse bias; the increase in quantum efficiency was at the $\sim 15\%$ level when the bias was increased to its maximum datasheet value. Two types of measurement were used to determine the increase in quantum efficiency as the bias was increased. The first method was to measure the DC response and the corresponding results are shown as circles. The second method involved modulating the bias and measuring the resulting modulation in the photocurrent, i.e. an AC measurement, and these are shown by the stars. Each set of measurements was done with a different intensity by changing either the beam's width or power.

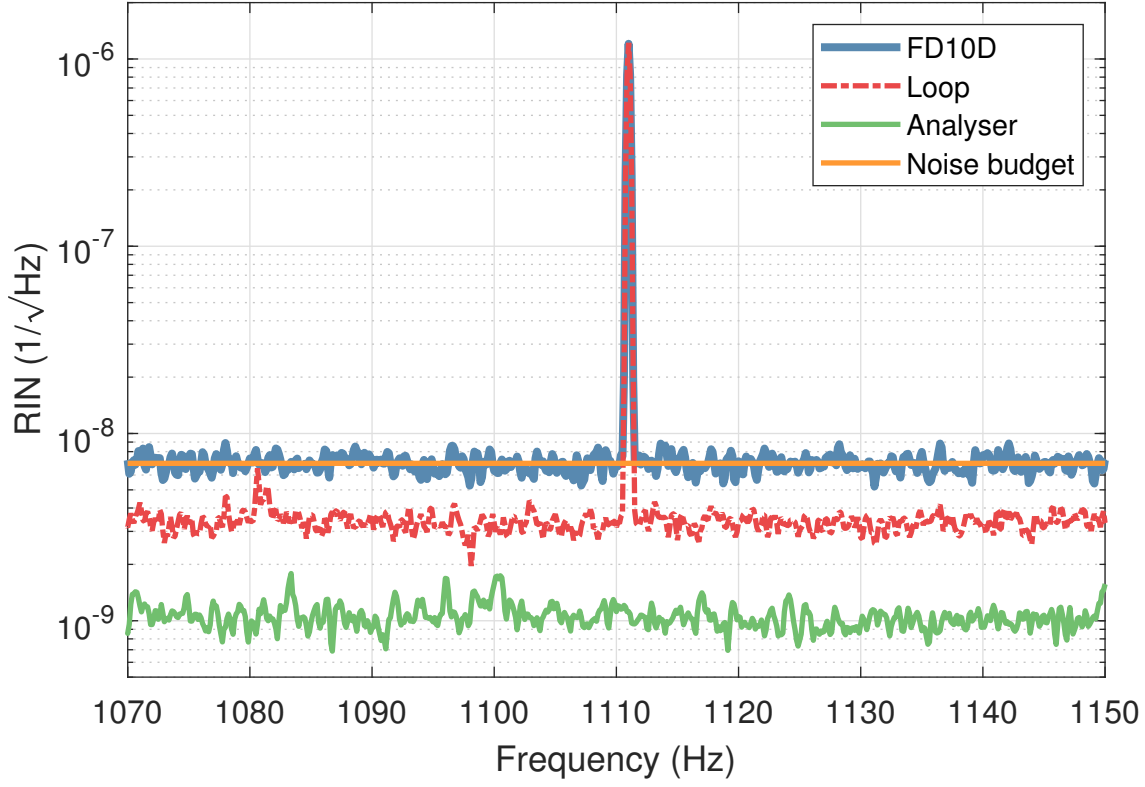


Figure 8.3: The bias of the FD10D was set to 0.6 V and it was exposed to 15 mW of 1550 nm light. 15 mA of photocurrent was generated. The noise of the photocurrent produced by the FD10D (blue line) corresponds to the level predicted by Equation 8.3 (orange line) meaning that that the quantum efficiency of the FD10D did increase rather than the increase in photocurrent being the result of an avalanche process. The 1111 Hz signal was used to calibrate the measurement.

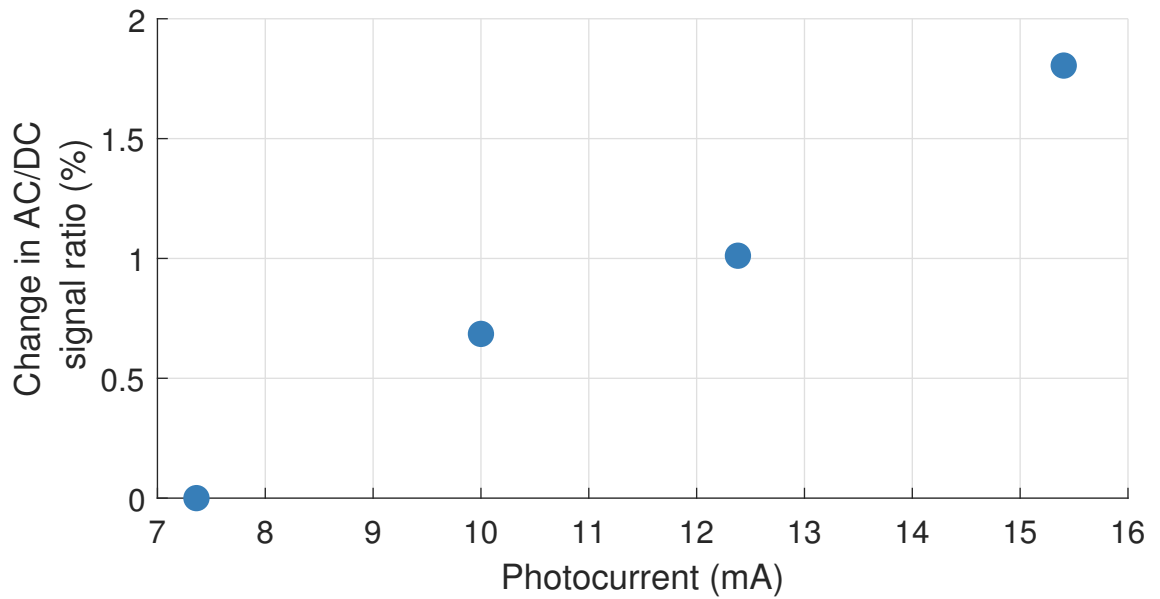


Figure 8.4: The bias of the FD10D was set to 0.6 V and it was exposed to a series of different power levels to check at what photocurrent the photodiode would saturate at. This was done by measuring the height of the calibration peak; if this peak decreased in size relative to the DC signal then this would indicate the photodiode was saturating. This peak could only be measured to within 1%, and so the FD10D showed no sign of saturation under these conditions.

8.5 Dark Current and Dark Noise in Extended InGaAs Photodiodes

The dark noise of the extended InGaAs photodiodes used in previous quantum optics experiments [123,124] resulted in the photodiodes being run without a bias, thus the dark noise of the photodiodes was a limiting factor in these experiments. The dark current and dark noise of extended InGaAs photodiodes depends on their reverse bias. The larger the reverse bias, the higher the dark current and dark noise.

There is a balance to strike between the quantum efficiency and dark noise. It is desirable to have as large a reverse bias as possible as this increases the quantum efficiency and saturation limit of the photodiode (see Figure 8.2). The dark noise of the photodiode should be below the shot noise level of the light that will be detected. For the reasons discussed in Section ??, the shot noise of a 10 mA photocurrent was used as a bench mark to compare the dark noise of the photodiodes at 10 Hz to.

To measure the dark noise produced by the extended InGaAs photodiodes, a transimpedance amplifier with high gain was used. This circuit, shown in Section ??, allowed for measurements of current noise several hundred times smaller than the shot noise of a 10 mA photocurrent ($6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$). The photodiodes were placed in a light tight box while the measurements were being made.

8.5.1 The Dependence of Dark Noise and Dark Current on Reverse Bias in Extended InGaAs Photodiodes

The dark noise as a function of bias was measured at room temperature ($\sim 20^\circ\text{C}$) for the photodiodes listed in Table 8.1 . The results for the FD10D are shown in Figure 8.5. The dark noise for the rest of the photodiodes can be found in Appendix ?? . Not all photodiodes had a simple noise spectrum; however, usually, the spectrum would fall as $1/f^{0.5}$.

The dark noise at 10 Hz and the dark current as a function of bias for each photodiode is shown in Figure 8.6. The noise at 10 Hz was either determined by reading the value from the corresponding graph in Appendix ?? or, if more precision was required, by fitting a $1/f^\beta$ slope to the data and inferring the noise from the fitted line.

When there were duplicate photodiodes to choose from, the superior photodiode was selected for characterisation. There can be a significant variation in quality within a batch of photodiodes. Figure 8.7 shows there was a difference of two orders of magnitude in the dark noise between two ‘identical’ Hamamatsu photodiodes.

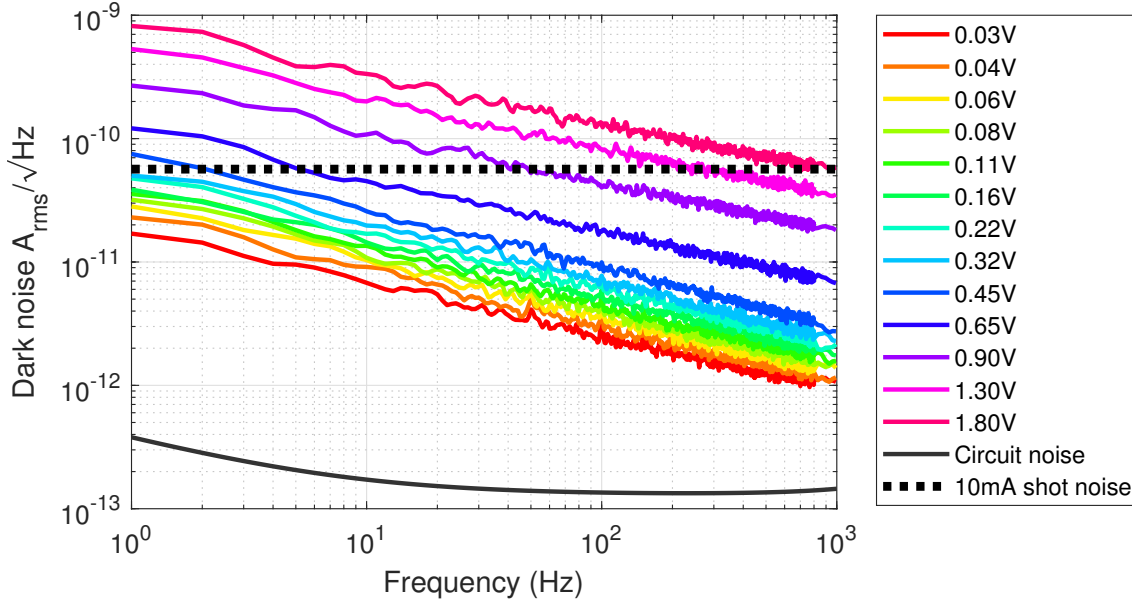


Figure 8.5: The dark noise of the FD10D increases as the reverse bias is increased. Above 0.6 V, the dark noise of the FD10D would limit the noise of a 10 mA current above 10 Hz. As the noise spectral densities have no knees, the function i_c/f^β can be fitted to the data. See appendix ?? for the dark noise spectra of other photodiodes. Below 0.9 V reverse bias, the spectra fall as $f^{-0.45}$ and above 0.9 V bias the spectra fall as $f^{-0.4}$. The values for β is typical for ‘1/f’ noise [138]. This data is shown in Figure 8.6 alongside the results for the other photodiodes tested.

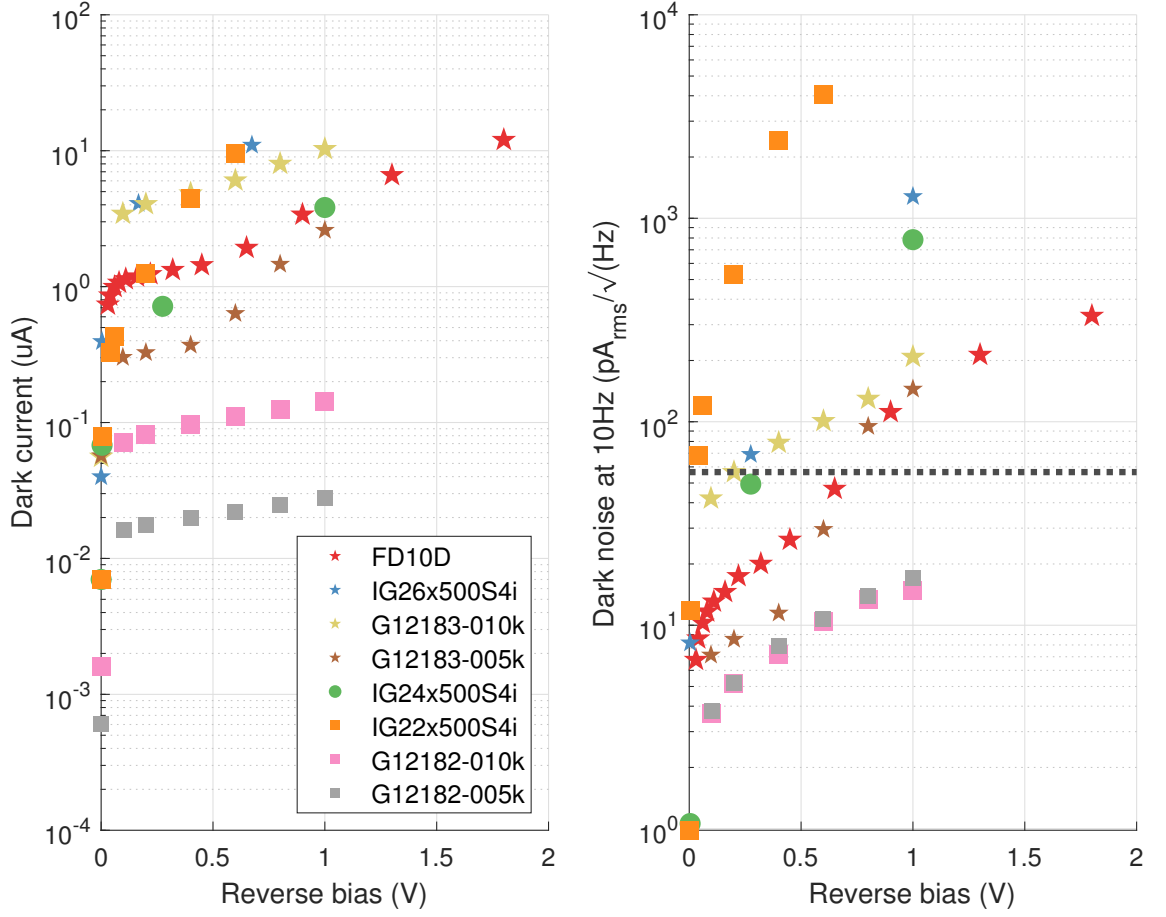


Figure 8.6: Left panel: the dark current for each photodiode was measured. Photodiodes of a similar cut-off wavelength share the same marker shape. The 1 mm photodiodes have larger markers than the 0.5 mm photodiodes. Right panel: the dark noise at 10 Hz for each photodiode. The dashed line indicates the level of shot noise for a 10 mA current.

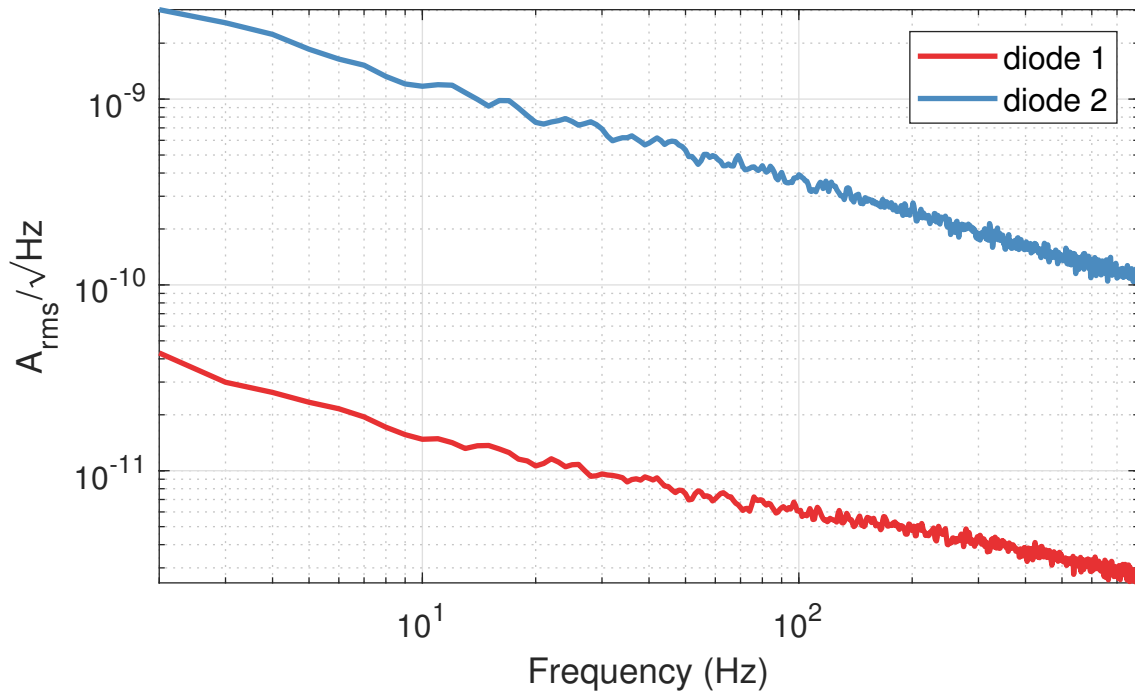


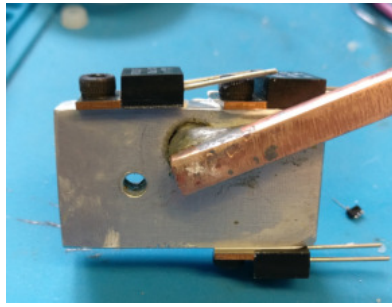
Figure 8.7: The difference in dark noise between two Hamamatsu photodiodes of the same kind (g12182-010k) under identical operating conditions (room temperature at 1 V bias). Photodiode 2 performs nearly 100 times better than photodiode 1.

8.5.2 The Dependence of Dark Noise and Dark Current on Temperature in Extended InGaAs Photodiodes

The dark noise of extended InGaAs photodiodes can be reduced by cooling them. The typical low-temperature damage threshold for extended InGaAs photodiodes is -40°C , so the dark noise of the photodiodes listed in Table 8.1 was measured at a range of temperatures from -40°C to 30°C .

The temperature of the photodiode was controlled with power resistors and an aluminium block thermally connected to a liquid nitrogen bath. The block is shown in Figure 8.8. This block had a copper rod soldered to it and this rod was inserted into the liquid nitrogen bath. A computer-controlled power supply (Temna 72-2540) was connected to the power resistors. A pt1000 temperature sensor and a multimeter (Keithley 2000), configured to make a four wire resistance measurement, were used to measure the temperature of the block. Labview was used to perform the PID calculation. This program read in the temperature of the block and calculated the voltage to set the power supply to.

The dark noise and the dark current for each photodiode in Table 8.1 was measured at the maximum reverse bias specified in its datasheet. The dark noise at 10 Hz and the dark current for each photodiode is shown in Figure 8.9. The individual dark noise spectra for each photodiode at each temperature is shown in Appendix ???. Except for the IG24x500S4i, the dark noise and dark current for each photodiode fell exponentially as the photodiode was cooled. The IG24x500S4i had a knee-like feature in its spectrum. Above 10 Hz, the dark noise fell as the IG24x500S4i was cooled, however below 10 Hz the noise spectra were bunched together and had a



(a)



(b)

Figure 8.8: (a) The photodiode was affixed to this aluminium block. The power resistors provided heating to the block and the copper rod provided a thermal link between the liquid nitrogen bath and the block. (b) A photodiode glued to the block such that a large proportion of the photodiode was in contact with it.

weak dependence on temperature.

8.6 The Saturation Limit of Some Laser Components Extended InGaAs Photodiodes

The Laser Components photodiodes were found to saturate at low powers if they were biased with low enough voltages such that their dark noise was below the noise target. The Laser Components photodiodes were exposed to a light modulation of known size. A similar type of measurement is reported in [146]. The size of the light modulation could be cross-checked by measuring it on the high performance in-loop photodiode. The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 8.10. A decrease in AC signal indicates that the photodiode is saturating.

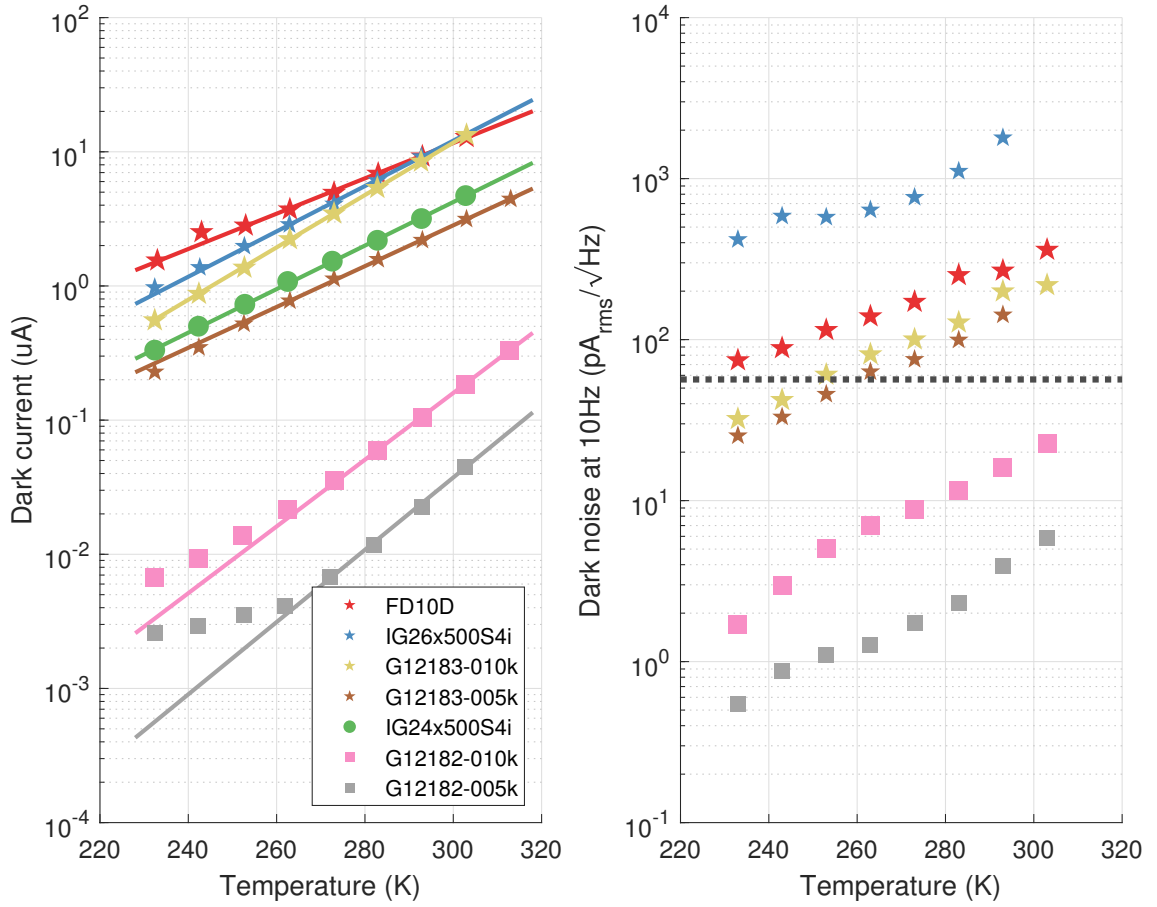


Figure 8.9: Left panel: the dark current for each photodiode at their maximum datasheet specified reverse bias as a function of temperature. The apparatus limited dark current measurements to a couple of nanoamps. Exponentials were fitted to the data. Photodiodes with similar cut-off wavelengths share marker shapes, and 1 mm photodiodes have larger markers than the 0.5 mm ones. The measurement of the DC dark current approached the noise floor of the apparatus around a few nanovolts. Right panel: the dark noise of each photodiode was measured as a function of temperature. The black dashed line indicates the shot noise level for a 10 mA current. At 10 Hz, the measurements for the IG24x500S4i did not change much as the temperature changed, there was a knee around this frequency (see Figure ??); the noise of the IG24x500S4i was $600 - 700 \text{ pA}/\sqrt{\text{Hz}}$ at 10 Hz.

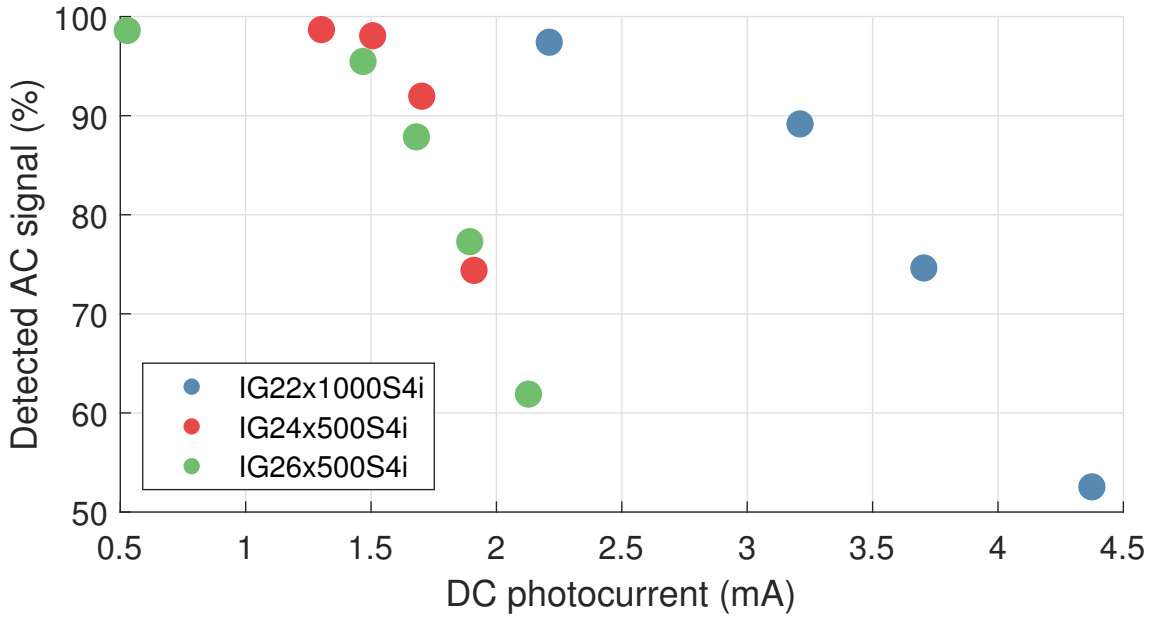


Figure 8.10: The saturation limit of three Laser Components photodiodes was measured. The photodiodes were biased so that their noise at 10 Hz was equal to the shot noise of a 10 mA current. The IG24x500S4i and IG26x500S4i were exposed to a beam whose 3σ width was 220 μm and the IG22x1000S4i was exposed to a beam whose 3σ width was 1000 μm . The size of the light modulation whose relative size was known was tracked as the DC photocurrent was increased. As the photodiodes saturated, the relative size of the AC signal decreased. These photodiodes had low saturation limits compared to the FD10D, which showed no signs of saturation up to 16.5 mA (see Figure 8.4).

8.7 Excess Noise in the Photocurrent made by the IG24x500S4i

The noise of the photocurrent produced by the IG24x500S4i exceeded shot noise by a factor of ~ 2 . This is shown in Figure 8.11. The set-up for this experiment is explained in Section 8.4. The photodiode was not saturating during this measurement as the 1111 Hz peak had the right relative intensity. This excess noise was found at 1.3 mA and 0.25 mA of photocurrent.

To rule out bias noise causing the excess noise, a noise projection measurement was made. The transfer function between the voltage at the anode of the photodiode and the output of the transimpedance amplifier was measured while 1 mA of photocurrent was being generated by the photodiode. A calibration modulation was applied to the bias voltage, and the noise at the anode of the photodiode and the noise at the output of the transimpedance amplifier were simultaneously measured. Using this method, 42.5 dB of bias voltage noise was removed from the measurement; however, this did not remove the excess broadband noise. This is shown in Figure 8.12.

The ideality factor of the IG24 was investigated. The ideality factor differing significantly from unity indicates that additional behaviour that was not described in Section 8.2 occurring within the photodiode. The equation of a photodiode is

$$I = I_p + I_s \left(\exp \left(-\frac{qV_b}{nk_B T} \right) - 1 \right), \quad (8.4)$$

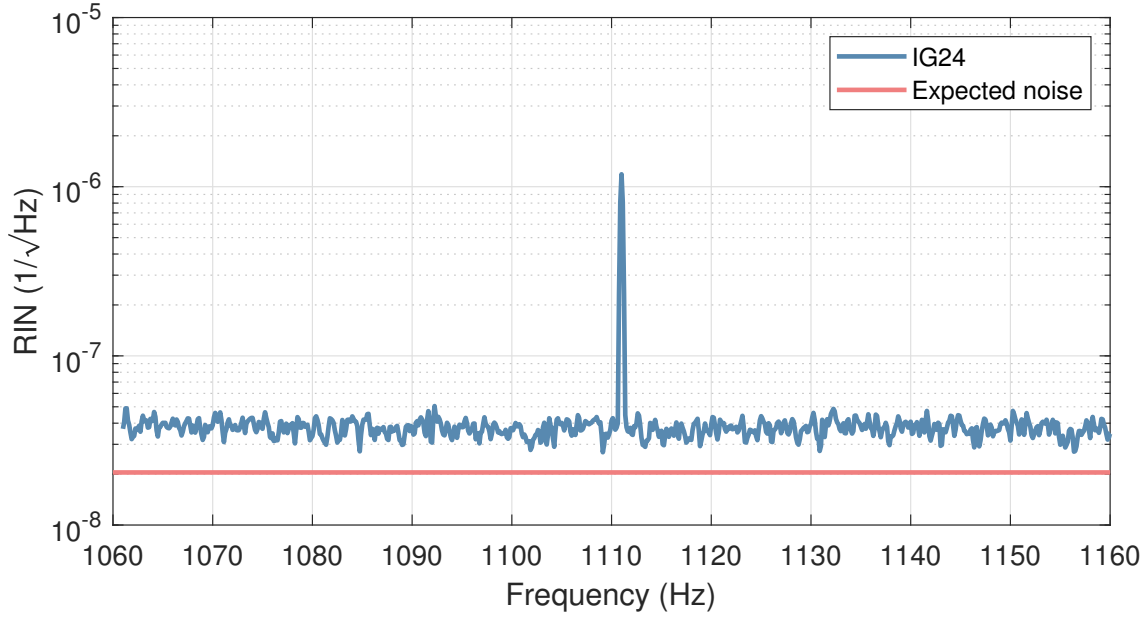


Figure 8.11: The noise of a 1.3 mA photocurrent made by the IG24x500S4i (blue) exceeded the expected amount (red) based on Equation 8.3 by a factor of 1.75. Note that the height of the 1111 Hz peak cannot be expressed in terms of amplitude spectral density units; however, this peak was measured to be the right level and so this beam was not saturating the photodiode.

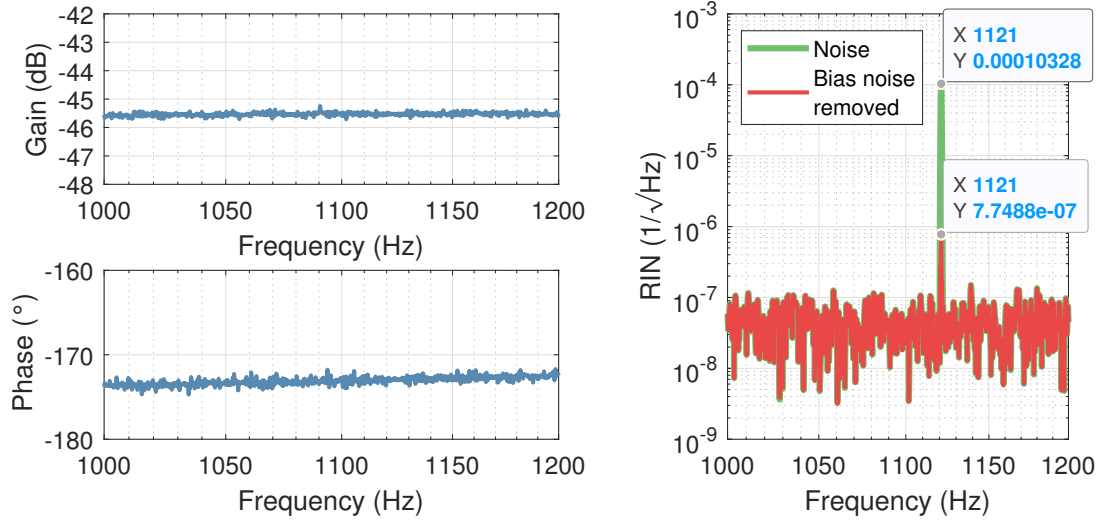


Figure 8.12: A noise projection measurement was performed to determine if the excess noise in the photocurrent produced by the IG24x500S4i (see Figure 8.11) was due to bias noise. The left hand panels show the bode plot of the transfer function between the photodiode's cathode and the output of the transimpedance amplifier when the photodiode was producing 1.3 mA of photocurrent. The right hand panel shows the effect of subtracting bias noise from the signal at the output of the transimpedance amplifier. The peak at 1121 Hz was a modulation applied to the photodiode's cathode; 42.5 dB of this signal was removed by using the noise projection technique which is in close agreement with the amount expected from the measured transfer function. However, the level of the broadband noise remained unchanged thus indicating that the excess noise witnessed in the photocurrent produced by the IG24x500S4i was not due to bias noise.

where I_p is the photocurrent, I_s is the saturation current of the photodiode, q is the charge of the electron, V_b is the bias of the photodiode, n is the ideality factor, k_B is Boltzmann's constant and T is the temperature of the photodiode. The current flowing through the photodiode as a function of forward bias was measured and is shown in Figure 8.13. The SciPy python function `scipy.optimize.curve_fit` was used to do an unweighted fit of the data to Equation 8.4. The photodiode was found to have $n \simeq 1$. An additional check of Equation 8.4 was made by shining light was shone onto the photodiode. This is shown in Figure 8.14.

The source of this excess noise is still unclear: the photodiode was not saturating as the calibration signal was the right level relative to the DC photocurrent, a noise projection measurement showed that the bias voltage noise was not the source of the excess photocurrent noise, and a measurement of the ideality factor was used to determine if the photodiode behaved as an ideal photodiode would. The level of noise cannot be explained by the Johnson-Nyquist noise of the shunt resistance of the photodiode as the excess noise is too high.

8.8 Properties of an InSb detector

InSb detectors have high quantum efficiency [126] to short infrared wavelength light. A Hamamatsu P5968 detector was tested, and the noise of the current produced by it was investigated. InSb detectors must be operated at low temperatures so liquid nitrogen was used to cool the photodiode to 77 K. The transimpedance amplifier used to measure the photocurrent is shown in Appendix ??.

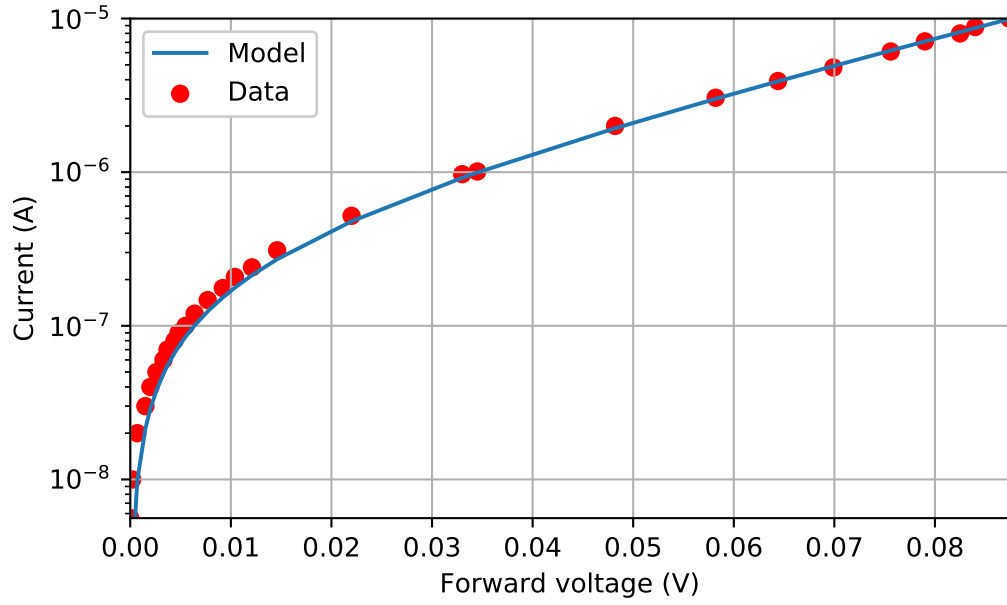


Figure 8.13: To investigate possible reasons why there was excess noise in the photocurrent produced by the IG24x500S4i, its ideality factor was measured (red points). Equation 8.4 was used to do an unweighted fit to the data ($I_0 = (360 \pm 10) \text{ nA}$ and $n = 1.02 \pm 0.01$). The IG24x500S4i was found to be close to ideal as n was close to unity and so this does not provide clues to the source of excess noise in the photocurrent.

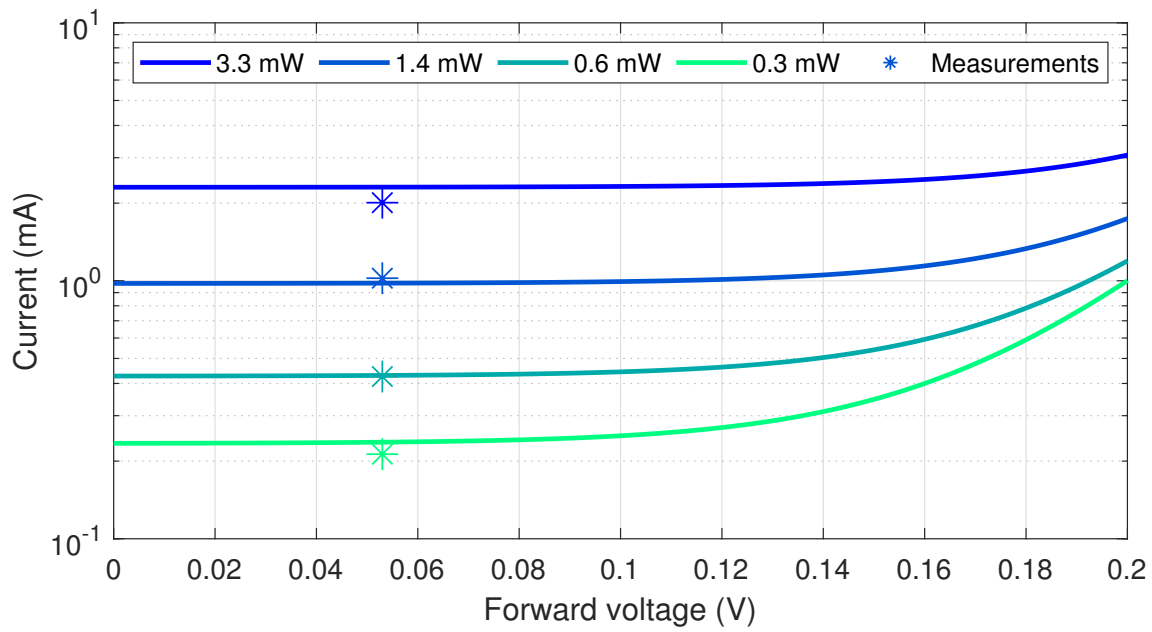


Figure 8.14: Different light levels were shone onto the photodiode and the current produced at 0.05 V reverse bias was recorded. These measurements are shown by the stars. Based on the model obtained from the data shown in Figure 8.13, the current flowing through the photodiode was predicted (indicated by the lines). The model and the measurements are in agreement with each other although the photodiode may have been saturating for the 3.3 mW measurement.

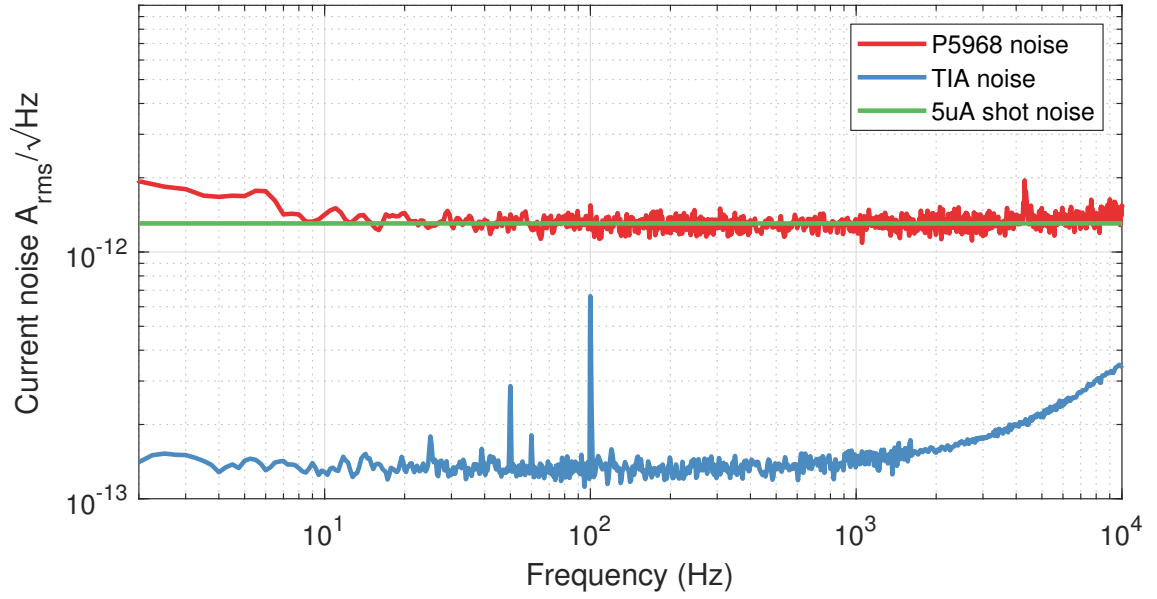


Figure 8.15: The P5968 InSb detector had its cathode grounded and the noise of the photocurrent due to the blackbody radiation produced by room temperature aluminium was measured (red). The expected level of photocurrent was calculated to be $5.5\text{ }\mu\text{A}$ and the measured photocurrent was $5.3\text{ }\mu\text{A}$. The noise level of the photocurrent corresponds to shot noise (green). The noise of the transimpedance amplifier (blue) is well below the shot noise of the photocurrent.

The detector was characterised by placing a low emissivity material in front of it. The InSb is sensitive to wavelengths of up to $5.3\text{ }\mu\text{m}$. Based on the solid angle covered by the material as seen by the detector, the expected level of thermal radiation that would be detected could be calculated. The expected photocurrent was $5.5\text{ }\mu\text{A}$. The DC current from the device was measured to be $5.3\text{ }\mu\text{A}$. The results from this test are shown in Figure 8.15. This corresponds to the device having a quantum efficiency of $\sim 90\%$.

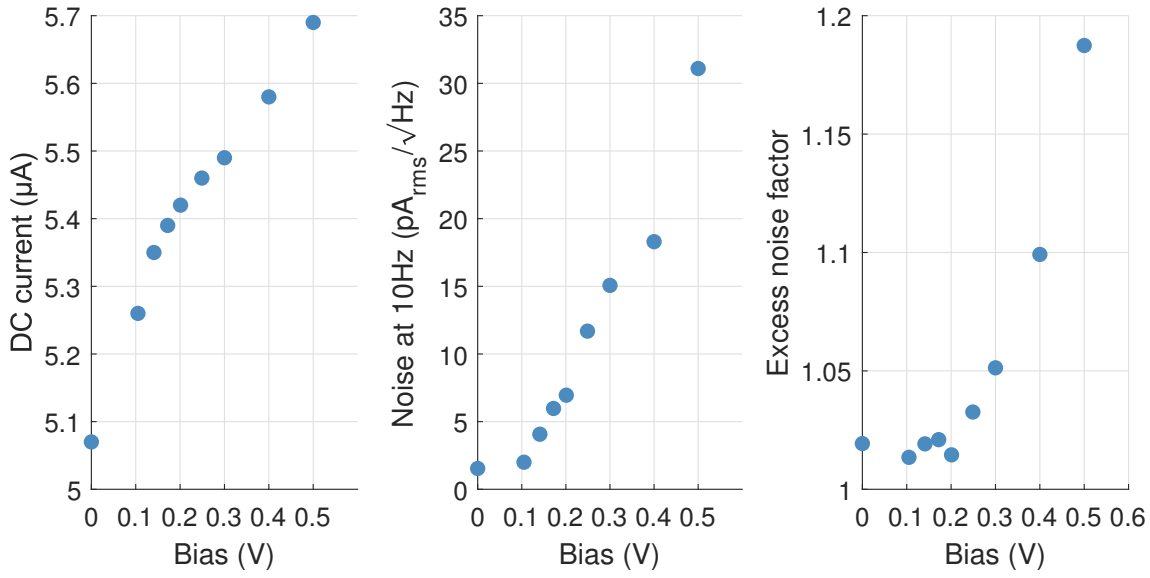


Figure 8.16: Left panel: the DC dark current of the P5968 as a function of bias. Middle panel: $1/f$ noise rose as the bias of the P5968 was increased (see Figure ??). Right panel: At a frequency where $1/f$ was not present, the noise of the photocurrent was measured. The excess noise factor is the multiplicative factor that the shot noise corresponding to the DC photocurrent needs to be multiplied by to get the noise level that was measured.

As the quantum efficiency of InSb photodetectors depends strongly on the reverse bias for $1.55\mu\text{m}$ wavelengths [147], the behaviour of the P5968 under a reverse bias was investigated. A reverse bias was applied to the device and this generated $1/f$ noise. The full data is shown in Section ???. The noise of the photocurrent also showed a broadband increase, this is shown in Figure 8.16.

8.9 Discussion

None of the photodiodes tested in this Chapter would meet the requirements described in Section 8.1 as the quantum efficiency is specified in all of their datasheets to be below 99%. Communication with several commercial manufacturers of extended InGaAs photodiodes suggest that there is no off-the-shelf photodiode capable of meeting the requirements set out in Section 8.1; the results in Section 8.4 and 8.5 support this and give insights into the best possible operating conditions for an off-the-shelf extended InGaAs photodiodes. As discussed in Section 8.2, high quality extended InGaAs photodiodes are difficult to fabricate because of the lattice mismatch between the InGaAs and the InP substrate.

The results cover seven different models of extended InGaAs photodiode and so it is possible that another photodiode manufacturer (e.g Teledyne Judson) have better performing photodiodes than the ones tested in this chapter. Additionally, as only two copies of most photodiodes were available for testing, batch-level statistics cannot be obtained from the results. For instance, the FD10D reported on in this chapter has lower dark noise than the G12183-010k; however, this does not tell one if FD10Ds are generally better than G12183-010ks.

Section 8.4 shows the effect that biasing an extended InGaAs photodiode can have on its quantum efficiency. The quantum efficiency of the FD10D was increased by 15% when its bias was set to its maximum value. This increase in quantum efficiency occurs because the electron-hole pairs that are excited by the incoming photons spend less time within the depletion region, thus they have a smaller chance of recombining with another hole/electron. This motivates using as large a bias as

possible as the quantum efficiency can be greatly increased.

The high amount of dark noise associated with the bias found in the Laser Components range of photodiodes (see Section 8.6) meant that they had to be operated with low biases if they were to be able to produce shot noise limited photocurrents above 10 Hz. This resulted in them having a low saturation level, see Figure 8.10. This is another reason why biasing a photodiode may be necessary. To avoid this saturation limit, one could split the beam and use an array of photodiodes to detect the light. However, better performing photodiodes could be used instead; the FD10D showed no signs of saturation, see Figure 8.4, when operating with a bias that fulfilled the noise criteria.

Applying a reverse bias to a photodiode reduces its capacitance, however a few hundred picofarads of capacitance that photodiodes have at low bias voltages is a manageable amount to design a transimpedance amplifier around for a DC readout scheme such as that used in the LIGO detectors. For instance, the photodiode circuit used in Section 8.4 (see Appendix ??) has a large enough bandwidth for a gravitational wave interferometer readout photodiode and was quiet enough to be limited by the shot noise of a 10 mA photocurrent.

While the dark current does not depend strongly on the reverse bias when the bias is below the tunnelling threshold, the dark noise of every photodiode increased as its reverse bias increased. Although photodiodes with higher dark current tended to have higher dark noise, the dark noise could not be predicted from the dark current. For instance, the FD10D had a larger dark current but lower noise than the IG24x500S4i. The dark current tended to be greater in larger photodiodes, this

is due to larger photodiodes being more likely to contain more defects. However, the Laser Components photodiodes break this rule as they were all of low quality. The Hamamatsu photodiodes with shorter cut-off wavelengths (less indium), had lower dark current and dark noise compared to the Hamamatsu photodiodes with longer cut-off wavelengths.

The dark noise can vary considerably amongst a batch of photodiodes, for example there was nearly a factor of 100 between the dark noise of the two Hamamatsu G12182-010ks that were tested (see Figure 8.7). Whether this is due to their manufacturing or how they were handled prior to purchase is unclear, however this result shows that it may be worth buying a batch of extended InGaAs photodiodes and selecting the best ones from it.

The dark noise of a photodiode can be significantly reduced by cooling it. The dark current and dark noise of a photodiode increases exponentially with temperature. Photodiodes can be purchased with thermoelectric coolers, however integration of such a photodiode into a vacuum system was not investigated and could pose an engineering challenge.

8.10 Conclusion

Extended InGaAs photodiodes are widely available detectors that can be used to sense $2\text{ }\mu\text{m}$ light; however, off-the-shelf detectors do not fulfil the requirements of a photodiode that can be used in the readout of a gravitational wave detector. This is due to gravitational wave detectors needing high quantum efficiency photodiodes so that the shot noise of the signal is not significantly enhanced.

It is technically challenging to create extended InGaAs photodiodes with high quantum efficiency due to the lattice mismatch between the InP substrate and the InGaAs. While the depletion region of a photodiode can be made larger by running the photodiode with a reverse bias, the dark noise of an extended InGaAs photodiode increases when reverse biased. Therefore, dark noise measurements of a selection of photodiodes (see Figure 8.9) were made to investigate suitable operating conditions for them.

To overcome the manufacturing challenges associated with creating extended InGaAs photodiodes, a wafer would need to be commissioned. Increases in quantum efficiency can be obtained by fabricating photodiodes to have anti-reflective coatings; however, the challenge of creating high quantum efficiency photodiodes lies in making it thick enough to absorb $> 99\%$ of the incident photons. Additionally, the photodiode must have a low amount of defects if it is to have a quantum efficiency $> 99\%$. Reference [148] shows results from a photodiode with 95% quantum efficiency and a cut-off wavelength of $2.4\text{ }\mu\text{m}$, however due to the exponential dependence of absorption on thickness, it is still a significant challenge to reach $> 99\%$ quantum efficiency. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a photodiode made from extended InGaAs can be reduced by orders of magnitudes by optimising this.

InSb detectors may also be useful as they have near unity quantum efficiency at short infra-red wavelengths, however their quantum efficiency below $2\text{ }\mu\text{m}$ can exceed unity due to multi-electron processes [126]. An InSb device was tested, however its

characteristics at $2\text{ }\mu\text{m}$ are still unclear. To investigate these, the device's window would need removing and a $2\text{ }\mu\text{m}$ laser would be needed.

Alternative photodiode materials could be used to detect $2\text{ }\mu\text{m}$ light. There have been suggestions that HgCdTe could be used to make a photodiode with $> 99\%$ quantum efficiency at $2\text{ }\mu\text{m}$ as HgCdTe does not suffer from strain related problems, however the linearity and dark noise of such HgCdTe photodiodes is unknown.

Appendix A

Extended InGaAs Diodes Noise Spectrums

Appendix B

Photodiode Circuits

Appendix C

FINESSE Simulations

Appendix D

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures D.2 and D.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [107] (figure D.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure D.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

D.0.1 Method

This section delineates the method described in section D and can be visualised using figure D.2.

Define the target mode at the optic 3. In figure D.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure D.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure D.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure D.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure D.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure D.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure D.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure D.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptres were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

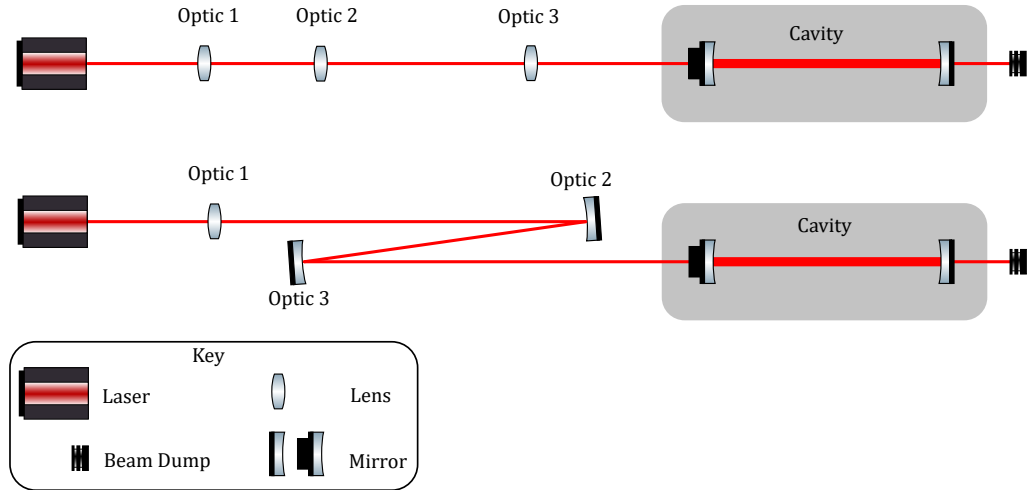


Figure D.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure D.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

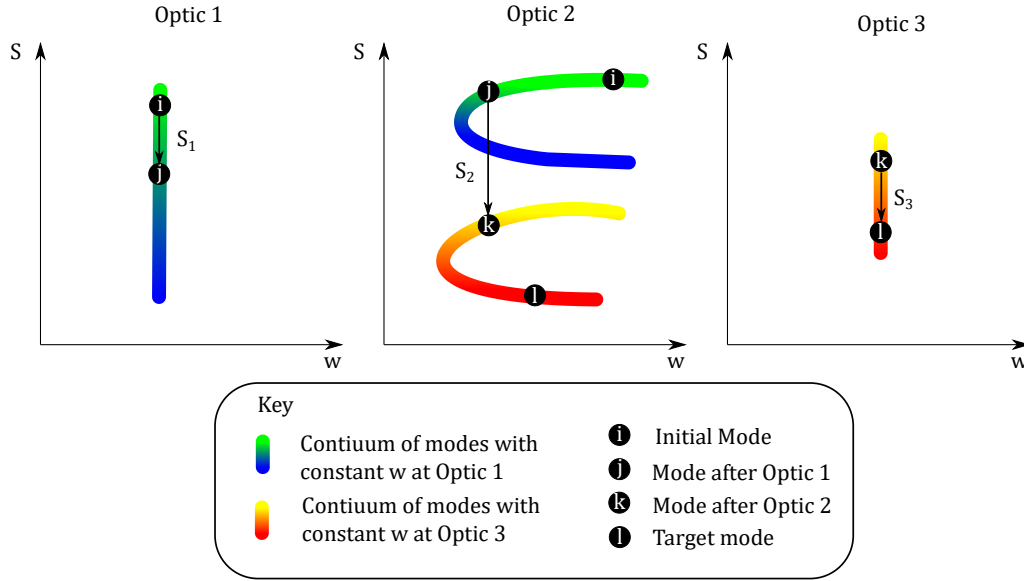


Figure D.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure D.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix E

Mode Matching Between 2 Astigmatic Beams

E.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{E.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{E.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{E.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{E.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{E.5})$$

The numerator of equation E.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{E.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{E.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation E.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{E.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{E.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{E.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{E.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - qx1^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{E.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{E.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}} \quad (\text{E.14})$$

allows equation E.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi(2q_{x1}^* q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi(2q_{x1} q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{E.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{E.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{E.17})$$

This means that the numerator of equation E.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{E.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation E.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{E.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term.} \quad (\text{E.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{E.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{E.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{E.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{E.24})$$

E.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{E.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{E.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{E.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{E.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{E.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

Appendix F

Light and Gaussian Modes

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